

Mind the Gap: Blockchain Protocols and Liquidity Mismatch *

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Abstract

Minable cryptocurrencies are designed so that new tokens are minted and transactions are processed when new blocks are confirmed and added to the blockchain. We hypothesize and test that, through a “delay” channel, block confirmations create swiftly changing mismatch in trading demand in the secondary market. Bitcoin data reveal dynamics of trading and returns consistent with this hypothesis. Tradable strategies exploring this effect generate large after-cost returns and exceptional Sharpe ratios, ruling out costs of arbitrage as an explanation. Bitcoin block confirmations also move the prices of other cryptocurrencies, suggesting strong non-fundamental price comovement in the cryptocurrency market.

Keywords: blockchain, Bitcoin, cryptocurrency, market microstructure, comovement

JEL Classification: G12, G14, D47.

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1 Introduction

The influence of cryptocurrencies has exceeded the expectations of many, with the total market capitalization surpassing \$3.4 trillion and the total trading volume surpassing \$200 billion per day as of Dec 31, 2024. This fast-growing market has attracted a vibrant literature that investigates issues revolving this market from the idea of decentralization with blockchain technology to the trading and prices of these new assets in the secondary market. Yet, the market consequences of some unique designs of these assets are underexplored by this young literature. In this paper, we investigate the market implications of the most basic feature of the blockchain protocols, namely block confirmations. Specifically, we ask a simple but nontrivial question: Does the original blockchain design induce frictions in the secondary market?

Our study is motivated by how transactions and new supply of minable cryptocurrencies are handled in the blockchain protocols. Taking Bitcoin as an example, the first blockchain invented by Satoshi Nakamoto in 2008, each Bitcoin block and the transactions included are confirmed by miners through solving a difficult mathematical problem before being added to the chain. This is also referred to as the proof-of-work process. The winning miners in these problem-solving contests are rewarded with not only a considerable amount of newly minted Bitcoins but also fees attached to the transactions in the block in the form of Bitcoins. This block and transaction confirmation is designed to occur every ten minutes on average although it is largely random from one block to another. The lengthy confirmations of Bitcoin have drawn an intensive debate with some applauding its better security while the others criticizing the delayed transaction confirmations.¹

In this paper, we investigate how these lengthy block confirmations affect the secondary market trading through their role of disciplining cryptocurrency flows. Hautsch et al. (2024) links the delayed confirmations to heightened risks for arbitragers, which lead to persistent price differences observed across secondary crypto markets. With a similar spirit, our study takes a different path by zooming in on whether and how delayed confirmations mechanically shape

¹Bitcoin's longer confirmation duration is considered a superior security feature as transactions tend to be deeper in the chain and thus more difficult to hack. However, critics argue that such a feature is only beneficial for large transactions and is unfavorable in general because of the delayed transaction processing and potential capacity issues. See a Wall Street Journal article for example: <https://www.wsj.com/articles/bitcoin-dodges-split-that-threatened-its-surging-price-1510172701>.

the trading behaviors in the secondary market. An extensive literature has documented the index rebalancing effects on stock prices using events such as S&P500 additions and deletions (Shleifer, 1986; Harris and Gurel, 1986), Toronto Stock Exchange index revision (Kaul et al., 2000), and Russell index reconstitution (Chang et al., 2015). Similar to these non-fundamental shocks, we believe that discrete block confirmations and exchange flows create a varying mismatch between buying and selling demand in the secondary market.

The specific mechanism is as follows. Cryptocurrencies are primarily trading on off-chain exchanges. When investors seek to sell their cryptocurrency stored in an on-chain wallet, they will first have to move the coins from this on-chain wallet to the exchange wallet. Like all other transactions, this transaction needs to be processed and confirmed on-chain and thus arrives in the exchange wallet at a time of block confirmation. Because all the exchange inflows are mechanically grouped together by block confirmations and released to the sellers on the exchange at the same time, the selling demand can be more clustered in the post-confirmation period. This mechanism is characterized in Figure 1 by the grey dots being delayed and becoming big red dots following the confirmations. In the meantime, buying cryptocurrency on an off-chain exchange is not subject to this complication and can occur instantly through the order books. Consequently, we expect a mechanically driven mismatch between the buying and selling demand that varies with the block confirmation cycle, with net selling immediately after confirmations and net buying in other times. We name this mechanism the “delay effect.”

In addition to this channel, the coin rewards that enter the wallets of miners may trigger their rebalancing demand and further contribute to the higher selling demand following block confirmations. Miners routinely sell coins to cover operational costs among other needs.² We name this channel the “reward effect.” However, the magnitude of this effect is dependent on the size of the reward relative to that of the block-level exchange flows. We discuss this potential effect in Section 3.2.

While exchange flows create noticeable selling demand, investors do not have to sell immediately following the block confirmation. However, for non-arbitragers, holding the position longer introduces additional risks that they may not be willing to bear. When this implicit cost

²The newly minted coins are usually subject to a lockup period before they are released from the on-chain wallet. For Bitcoin, for example, this period is 100 blocks. Yet, it is reasonable to assume that professional miners maintain a tradable balance in an exchange wallet to avoid delaying their demand to sell.

is higher than the cost associated with the price pressure, investors intuitively trade off potentially better selling prices for faster liquidation. The random nature of the block confirmations also introduces additional uncertainty about the short-term demand-supply dynamics, further increasing the implicit holding cost.³ That said, how investors behave in this case is eventually an empirical question.

This feature of blockchain technology is shared by all minable (proof-of-work) blockchain assets, with the only difference being the designed target wait time between block confirmations. Across the most popular minable cryptocurrencies, the target wait time ranges from 1 minute (DOGE) to 10 minutes (Bitcoin). We conduct our primary analyses using Bitcoin blockchain and trading data for three reasons. First, Bitcoin is the most popular minable cryptocurrency with the total market capitalization of about \$1.85 trillion as of Dec 31, 2024. Second, as the first and original blockchain asset, Bitcoin has better liquidity and market depth compared to its later peers. Third, Bitcoin has a target wait time of 10 minutes, the longest among all cryptocurrencies, and thus better facilitates our minute-by-minute investigation of the confirmation effect.⁴

Occurrences of Bitcoin block confirmations are largely random, with the block-to-block duration ranging from 1 minute for the 10th percentile to 23 minutes for the 90 percentile in our sample. The average is close to the designed length of 10 minutes (Nakamoto, 2008). We investigate our hypotheses using data from Binance, the largest and most liquid Bitcoin market, which helps us make general inferences. The summary statistics provide initial evidence against the “reward effect” being a substantial factor given that the average block reward is quite small compared to the exchange inflow per block. Our estimated winning miners’ flow to Binance accounts for far below 1% of the average exchange inflow. That said, consistent with our “delay” hypothesis, we find that market-order selling significantly increases by a rate ranging from 3.1% to 6.5% in the five minutes after the confirmation compared to the pre-confirmation level. This increase in selling demand drives the market order imbalance from slightly positive

³While the difficulty level of the mathematical problem governs the expected time for solving the problem, the actual time used from block to block is largely random. This is similar to the idea that time required for solving a complex maze can be substantially shorter if the player accidentally makes all the right turns.

⁴The newer altcoins in general tend to have much shorter target wait time to accommodate the transaction speed. But some of them, such as Bitcoin Cash and Litecoin, keep relatively long wait times for security considerations.

in the pre-confirmation period to -2.73% after the confirmation.

This periodic selling demand following block confirmations substantially moves Bitcoin prices. Although the overall return of Bitcoin is positive due to the increase in Bitcoin value in our sample period, the minute-level returns in the post-confirmation period are significantly negative, ranging from -0.274 to -0.099 bps in the first five minutes after confirmation. These results are robust to regressions controlling for time fixed effects. These trading and return patterns also exist in the futures market, where the magnitude of the effect is comparable to the spot market. This is likely due to arbitrage activities between the spot and futures markets, since futures trading does not require actual bitcoin flows.

Our “delay” hypothesis stems from the fact that there is noticeable wait time between blocks. As such, this wait time clearly plays an important role in the pattern and the magnitude of the price variations. Specifically, the wait time before a successful confirmation determines how many transactions are delayed and accumulated in the block. In doing so, the wait time provides three testable implications. First, a long-awaited confirmation groups together more Bitcoin flows to the exchange, creating a larger mismatch between buying and selling in the post-confirmation period, i.e., a stronger “delay effect.” Second, if the wait time and transactions processed have a concave relationship, then we should observe stronger selling pressure when a larger total flow was brought in by two block closely positioned block confirmations. Third, as the negative price pressure runs out of steam, we expect the price to reverse. But after the price fully reversed, the “delay effect” may keep driving the price higher as an increasing amount of selling demand is withheld by the upcoming block confirmation, which starts a new block cycle.

Supporting these predictions, we indeed find that both long-awaited and overlapped block confirmations both have significantly larger impacts on subsequent order imbalances and returns, with the former being particularly strong. The latter signifies the confirmation effect due to a concave relationship between the wait time and the number of transactions processed by the block. Following through to the subsequent block confirmation, we find that the price drop not only fully reverses, but also turns to a substantial increase if the upcoming block confirmation occurs sufficiently late.

We complement the wait-time analyses with tests based on block-level exchange flows.

Recent studies, including Griffin and Shams (2020) and Makarov and Schoar (2025), use clustering algorithms to identify cryptocurrency wallets that belong to certain exchanges and study inter-exchange coin flows. We directly characterize block confirmations using their associated Bitcoin flows as a proxy for the selling demand and conduct a heterogeneity test across blocks. Consistent with our hypothesis, controlling for wait time, block confirmations with large exchange inflows more than double the negative returns in the post-confirmation window.

Next, we delve into the magnitude of this confirmation-induced price variation by separating minute-level observations in our sample into the post-confirmation periods, defined as the first five minutes following the most recent confirmation, and other periods. Avoiding post-confirmation periods results in a highly significant average daily return of 0.827%. Holding short positions only in the post-confirmation periods results in a highly significant average daily return of 0.672%. These strategy returns are compounded to an average daily return of 1.515% compared to the average daily return of 0.139% for the passive buy-and-hold strategy, which is only marginally statistically significant. The reward-halving events have virtually no impact on the strategy returns, confirming the insignificance of the “reward effect.”

Bitcoin trading has developed a reputation of high liquidity and small bid-ask spreads in recent years. In our full sample beginning 2018, the long and short strategies combined still generates a statistically significant average daily return of 0.204% after transaction costs. This number ranges from 0.749% to 1.691% from 2020 onward where the bid-ask spreads dropped to below one basis point, with the highest return observed in 2024, the last year of our sample. During this period, this strategy also generates exceptional after-cost Sharpe ratios ranging from 3.684 to 10.322.

Finally, we examine whether the block confirmation effect of Bitcoin, the largest cryptocurrency, influences trading and prices of other popular cryptocurrencies, including Ethereum, BNB, XRP, and Solana. The previous studies, such as Hu et al. (2018), Liu and Tsyvinski (2021), and Liu et al. (2022), document price comovement of cryptocurrencies in the secondary market, which is attributable to fundamental factors related to the cryptocurrency’s network and production costs. However, the block confirmation effect documented in this study is clearly transient and non-fundamental, which should not cause market-wide comovement in an efficient market. In sharp contrast to this logic, we find that the Bitcoin block confirmation effect

significantly influences the trading and prices of the altcoins in the same direction as Bitcoin. This result further questions the efficiency and how investors trade in this market.

2 Literature and Contribution

Our study is aimed at promoting a better understanding of the design of cryptocurrencies based on the novel blockchain technology. There are two strands of literature studying the cryptocurrency design. The first strand of literature tries to understand the pros and cons of such design in the intended role of cryptocurrencies, namely the transaction processing. These studies include Cong and He (2019), Huberman et al. (2021), Saleh (2021), Sokolov (2021), Easley et al. (2019), Cong et al. (2021a), Cong and He (2019); Capponi and Jia (2021); Li and Mann (2018); Lee et al. (2021); Hinzen et al. (2021); Easley et al. (2019); Hasbrouck et al. (2025); Biais et al. (2019); Schilling and Uhlig (2019); Sockin and Xiong (2023) among others (See Harvey et al. (2021); John et al. (2022, 2023) for reviews.). A recent study by Makarov and Schoar (2025), however, shows that the vast majority of the crypto-transactions are financial transactions in nature. Indeed, the liquidity of the secondary cryptocurrency market has improved substantially in the recent years. A second strand of literature thus focuses on the secondary-market impacts of the cryptocurrency design and the blockchain technology (Pagnotta, 2022; Cong et al., 2021b; Liu and Tsyvinski, 2021; Biais et al., 2019; Capponi and Jia, 2021). Liu and Tsyvinski (2021) construct network and production factors of cryptocurrencies investigate how they are related to returns. Cong et al. (2021b) study how token platforms influence equilibrium token prices. Pagnotta (2022) studies how the Bitcoin design could lead to high price volatility unrelated to fundamentals. Biais et al. (2023) investigate the Bitcoin return variations attributable to its fundamental transactional benefits versus extrinsic volatility. These studies highlight the significant market and liquidity risks inherent in digital assets, but the specific frictions caused by the protocol's supply mechanism remain underexplored. Our study contributes to this second strand of literature by documenting a large confirmation effect in the cryptocurrency market. This effect increases in the wait time between blocks, suggesting that newer cryptocurrencies with seconds or fractions of a second between blocks are less prone to this effect. Our findings also advocate the recent development and utilization of Layer-2 networks that help solve issues

with the original blockchain design including the long wait time.⁵

Our study also provides important insights into the quality of the cryptocurrency market. Previous studies document evidence that the market efficiency for cryptocurrency is poor. Griffin and Shams (2020) find that Bitcoin price movements are associated with the issuances of the most commonly used intermediary stablecoin Tether. Makarov and Schoar (2020) show that large arbitrage opportunities exist across cryptocurrency exchanges. Hautsch et al. (2024) attribute this lack of cross-exchange arbitrage to delayed settlement or block confirmations. Using data from the largest and most liquid cryptocurrency exchange Binance, we zoom in on the directional effect of the block confirmations on prices in the secondary market. These results echo the fact that cryptocurrency market is dominated by retail investors who do not actively trade or rebalance (see (Kogan et al., 2023)), which potentially leads to less efficient pricing.

Finally, our results have implications for comovements and common factors in the cross-section of cryptocurrency market and beyond (see, for example, (Liu et al., 2022; Liu and Tsyvinski, 2021; Hu et al., 2018; Corbet et al., 2018; Lustig et al., 2011)). Adding to this literature, we document evidence in an event-study setting that non-fundamental variations of Bitcoin prices lead to price variations in other cryptocurrencies in the same direction. This suggests that the risk factors that capture comovement in this market are capturing not only effects with fundamental and/or market-wide implications but also idiosyncratic effects that are transient and non-fundamental.

3 Data

3.1 Sample

To construct our sample, we first obtain the Bitcoin block-level information from Blockchair. This dataset covers information including block height, hash, timestamp, size/weight, number of transactions, total fees, block reward, and difficulty of the Bitcoin blocks. We provide more institutional details related to block confirmations and mining rewards in the Appendix. We

⁵For example, Binance integrated the Bitcoin Lightning Network in 2023 to help facilitate Bitcoin deposits and withdrawals without on-chain restrictions. However, we still observe a substantial confirmation effect in 2024, as shown in Figure 7. This may be attributed to its low adoption rate by investors owing to concerns about the security and additional complexity, which could change over time.

also obtain the millisecond-level Bitcoin trading and price data from Binance, the largest and most liquid cryptocurrency exchange that accounts for roughly 40% of the total crypto-trading volume. Specifically, we use BTCUSDT aggregated trade data (aggTrades) and minute-level candlestick data (klines) to calculate the variables below. Bitcoin is quoted in a variety of fiat currencies and stablecoins. But the some fiat currencies (e.g. USD) can only be used to convert to cryptocurrencies where the exchange is the counterpart and charges a higher fee. For the actual market trading, investors mostly use BTC-stablecoin pairs. The most liquid pair during our sample period is the BTCUSDT, which is Bitcoin quoted in Tether (USDT). We use the spot market information of BTCUSDT for our baseline results. This dataset includes information on unique aggregated trade identifiers (trade_id), the execution price (price), the total quantity traded (quantity), the millisecond timestamp (timestamp), and a boolean aggressor flag (isBuyerMaker), along with minute-level open, high, low, close, and volume from klines. With this data, we construct minute-level returns (close-to-close), market-order buy and sell volumes, order imbalance, and bid-ask spread in percentage. In addition to these two primary datasets, we also obtain the block-level inflows and outflows to Binance from one of the leading crypto analytic platforms, CryptoQuant, who identifies exchange wallets using algorithm similar to Griffin and Shams (2020) and Makarov and Schoar (2025).⁶

3.2 Summary Statistics and Random Block Confirmations

We report the summary statistics of our sample in Table 1. Our sample encompasses a total of 375,163 blocks. At the block level, the average wait time between blocks is 9.815 minutes, very close to the designed target wait time of 10 minutes. That said, the actual occurrences of block confirmations vary drastically from 1 minute for the 10th percentile to 23 minutes for the 90 percentile with a median of 7 minutes, suggesting positive skewness. The number of transactions confirmed in each block also varies from 592 at the 10th percentile to 2,919 at the 90 percentile. This variable is naturally correlated with the wait time before the confirmation. A longer wait time allows more transactions to be included in a block. The total transaction fee has a mean of 0.366 Bitcoin and a median of 0.167 Bitcoin. This variable is also positively

⁶CryptoQuant uses a clustering technique to identify exchange wallets similar to Griffin and Shams (2020) and Makarov and Schoar (2025). The detailed procedures are described in CryptoQuant (2021).

skewed and highly correlated with the wait time through the number of transactions included in the block. These values are smaller but nontrivial compared to the newly minted Bitcoins per block. Our sample encompasses two Bitcoin halving events. The first halving occurred on May 11th, 2020, reducing the Bitcoin minting from 12.5 to 6.25 coins per block. The second and most recent halving was on April 19th, 2024, further reducing the minting to 3.125 coins per block. We discuss the impacts of these halving events in Section 5.1. As the total reward to the miners is the sum of the newly minted coins and the transaction fees, with the former decreases over time, the latter is becoming an increasingly important part of the total reward. Comparing to the exchange inflows and outflows, which are on average 84 and 81 coins per block, the block rewards are quite small. In unreported results, we calculated the average winning miners' flow to Binance as a percentage of the total Binance inflow. It is far below 1% and thus evident against the “reward effect” being substantial. It is also noticeable that exchange flows show substantial variations across blocks. We confirmed in the data that these large variations are not simply due to a time trend. The bottom section of Table 1 summarizes the minute-level trading and return variables.

Figure 2 depicts the random nature of Bitcoin block confirmations. The heat map of confirmations across a typical day and over a three-month period reveals no intraday or cross-day patterns. We randomly selected Dec 12, 2021 for the intraday pattern and October to December 2021 for the cross-day pattern as they are the closest to the end of our sample period. Within the day in Panel (a), a 30-minute bin could contain as many as 7 or as few as 0 block confirmations. Panel (b) further shows that there are no clear day-of-the-week or day-of-the-month effects among block confirmations. Serial correlations of wait time are not reported in this paper but are all close to zero. This random nature introduces additional uncertainty regarding the short-term demand-supply dynamics, incentivizing the winning miners to liquidate immediately. Additionally, high-frequency investors are not able to predict and front-run these confirmations.

4 Block Confirmation Effect

These discrete block confirmations not only create temporary selling demand from winning miners (the “reward effect”) but also interrupt the continuous Bitcoin flow into the secondary market (the “delay effect”). Our hypothesis predicts that both forces repeatedly induce mismatched buying and selling demand at a high-frequency level. Whether and to what extent these forces move Bitcoin prices is an empirical question and potentially reveals the quality of this market. Previous studies, such as Griffin and Shams (2020) and Makarov and Schoar (2020), challenge the efficiency of the crypto-market with evidence of price manipulation and cross-exchange arbitrage opportunities.

In this section, we empirically investigate the market consequences of these shocks. We begin by examining high-frequency trading dynamics and price behavior in the immediate aftermath of block confirmations. We then delve into the economic mechanisms that drive these patterns, focusing on how the time between blocks and exchange flows influence the return dynamics.

4.1 Bitcoin Trading around Block Confirmations

Do block confirmations induce temporarily mismatched buying and selling demand in the market? To investigate this question, we first analyze the trading activities in a narrow window surrounding block confirmations. With details of Bitcoin transactions on Binance, we construct several variables to gauge market trading at minute-level frequency. The market buy volume is the total volume involving buying using market orders. Similarly, the market sell volume is the total volume involving selling using market orders. The sum of these two volumes gives the total trading volume in that minute. From these two variables, we also calculate the market order imbalance (*ordimb*) as follows:

$$Ordimb_t = \frac{market_buy_volume_t - market_sell_volume_t}{market_buy_volume_t + market_sell_volume_t}. \quad (1)$$

If block confirmations indeed induce abnormal selling, we expect the market sell volume to increase and the order imbalance to decrease following these events.

Figure 3 plots the averages of these four variables across all block confirmations, from eight minutes before to eight minutes after the confirmation. Panel (a) and (b) demonstrate that both the market buy and market sell volumes increase immediately after the block confirmation, with the latter showing a much more pronounced response to the event than the former. In the eight minutes before the confirmation, the market sell volume revolves closely around 25.2, with an overall average of 25.255. This number increases to 25.511 at $t = 1$ and 25.800 at $t = 2$ before reaching the peak of 26.031 at $t = 3$, and then gradually reverses afterward. The average market sell volume in the five-minute post-confirmation window is 25.735, 1.9% higher than the pre-confirmation window average. In contrast, the market buy volume experiences a much milder increase of about 0.101 (0.4%) from before to after the confirmation, which could be induced by potentially more aggressively priced limit sell orders. Unsurprisingly, the asymmetric patterns in market buy and market sell result in a sharp drop of the market order imbalance, shown in Panel (c), from near zero to as negative as -2.1% at $t = 3$ before gradually reversing.

4.2 Bitcoin Returns around Block Confirmations

Whether this surge in sell-side order flows moves Bitcoin prices speaks directly to the depth and quality of the secondary crypto-market. Similar to Figure 3, we begin this investigation by plotting the average minute-level Bitcoin returns around block confirmations. In addition to a simple return based on the closing (last) price of each minute, we also utilize a return measure based on the midpoint of the bid and ask prices to help mitigate the impact of bid-ask bounce. Due to the lack of detailed order book data, we use the last pair of transactions executed on the bid and ask prices to proxy for the bid and ask of the minute, respectively. In rare cases where the last bid is higher than the last ask, we use the last transaction price as the bid-ask midpoint and assume the spread is zero.⁷

In Figure 4, we plot the average returns across all block confirmations, from eight minutes before to eight minutes after the confirmation. Panel (a) uses the closing-price return. In the eight minutes before the confirmation, the minute-level returns range between -0.004 bps and 0.046 bps, almost all insignificantly different from zero. The confirmation clearly induces

⁷The bid-ask spread of Bitcoin trading on Binance is typically small, which we show in Figure 8. We include this measure in our baseline tests to further mitigate the bid-ask bounce.

significantly negative returns in the five minutes immediately after, ranging from -0.207 bps $t = 3$ to -0.056 ($t = 5$). This pattern resembles that of the order imbalance documented above, indicating a close connection between the order imbalance and the return. For example, the negative order imbalance of -2.110% at $t = 3$ in Figure 4 is translated into a significantly negative return of -0.207 bps here. These negative returns in the five-minute post-confirmation window add up to a return of -0.651 bps per block confirmation. We discuss the economic meaning of this magnitude in Section 5. The bid-ask midpoint return in Panel (b) confirms this pattern with little difference in economic magnitude or statistical significance, ruling out the bid-ask bounce as an explanation.

To further probe the breadth of the confirmation effect, we examine its presence in the Bitcoin futures market. The result supports that arbitrage activities carry this effect from the spot market to the futures market. Figure 5 plots the minute-level order imbalance and return of Bitcoin futures around block confirmations. Here, we focus on the perpetual futures, which are the most liquid contracts. Similar to the patterns documented in Figures 3 and 4, Panel (a) and Panel (b) of Figure 5 clearly demonstrate negative order imbalances and returns immediately after block confirmations, respectively. The magnitudes are comparable to those in the spot market. The existence of this pattern in the futures market is likely due to arbitrage activities between the spot and futures markets, since futures trading does not require actual bitcoin flows.

4.3 Robustness of the Confirmation Effect

While the visual patterns are simple and intriguing, they gloss over some empirical complications, such as potentially overlapped event windows and potential time clusters. To formally quantify these visual patterns, we employ a regression framework. Specifically, we link each minute-level trading or return variable to variables indicating where this minute is positioned relative to the most recent block confirmation. The specification is as follows:

$$Y_t = \alpha + \sum_{\tau=1}^5 \gamma_{\tau} \text{Post}_{\tau,t} + \delta_{Reset} + \zeta_{DOW} + \eta_{HOD} + \epsilon_t. \quad (2)$$

Here, Y_t is the trading or return variable of interest at minute t . We consider trading variables including the natural log of market sell volume (LogSell), the natural log of market buy vol-

ume (LogBuy), the order imbalance (Ordimb), and the percentage bid-ask spread. For return variables, we consider the spot market returns based on the closing prices and bid-ask midpoints. $Post_{\tau,t}$ is an indicator that equals one if minute t is τ minutes after the most recent block confirmation, and zero otherwise. We also include reset fixed effects (δ_{Reset}) as controls for slow-moving time trends. A difficulty reset adjusts the difficulty level of the mathematical problem to the changing computing power from miners. This reset occurs every 2,016 blocks to maintain a 10-minute target wait time between blocks. The reset fixed effects help mitigate the concern that our results above are concentrated in early part of our sample where liquidity is relatively low for Bitcoin trading. We also include day-of-the-week (ζ_{DOW}) and hour-of-the-day (η_{HOD}) fixed effects to control for the interday and intraday seasonalities. If the patterns above are robust, we expect the coefficients γ_{τ} , which capture the post-confirmation effect, to be significantly different from zero.

Table 2 presents the regression estimates where the dependent variables are trading measures. Column (2) shows that the natural log of market sell volume is significantly higher in all first five minutes after the confirmation. For example, the coefficient on $Post3$ indicates that the market sell volume is approximately 6.4% higher in the third minute after a block confirmation (t-stat=16.78). In comparison, the coefficients for log market buy volume in column (1) are smaller and less significant. This sell-side dominance is reflected in the market order imbalance ($Ordimb$) in column (3), where the coefficients are large, negative, and highly statistically significant. The magnitudes of these coefficients indicate 1.020% to 2.732% lower (more negative) order imbalances in this five-minute window, with the most negative order imbalance observed at the third minute. The coefficients in column (4) show a statistically significant widening of the bid-ask spread by approximately one basis point, peaking three minutes after the event, which is consistent with liquidity providers widening spreads in response to the directional order flow.

The return regressions are reported in Table 3. Column (1) and (2) use closing-price return and bid-ask midpoint return, respectively. In both columns, the results show statistically and economically significant negative returns in each of the first five minutes. In the column (1), for example, the magnitudes of these coefficients compound to a total lower return of 0.977 bps following a confirmation. The results in column (2) are quantitatively similar. The small values

of the R-squared suggest that the Bitcoin returns are highly volatile within each event minute captured by the indicators. This poses a potential challenge to exploiting this pattern with a tradable strategy. We explicitly investigate the investment implications of this result in Section 5.

Overall, these regression results confirm the patterns in Figures 3 and 4, reinforcing the conclusion that block confirmations trigger a genuine, albeit temporary, price drop driven by selling pressure. These results are not specific to a short period of our sample.⁸

4.4 The “Delay” channel: Evidence from Wait Time between Blocks

The average patterns and their magnitudes so far provide clear support to our hypothesis. But the actual return dynamics vary from one block to another, with the most obvious differentiating point being the inter-block duration. In this section, we take an additional step to explore how the wait time between blocks influences the return dynamics and attempt to gain insights into the proposed channels.

Specifically, we test three implications of our “delay” hypothesis. First, a long-awaited confirmation groups together more Bitcoin flows to the exchange, creating a larger mismatch between buying and selling in the post-confirmation period, i.e., a stronger “delay effect.”⁹ Second, the wait time and the number of transactions processed have a positive and clearly concave relationship as shown in Figure 6. A block confirmation with near-zero wait time still processes 1,123 transactions as compared to the 1,627 transactions of a 11-to-15-minute block. Therefore, we expect to observe stronger selling pressure following two closely positioned blocks due to a larger total coin flow. Third, because the abnormal selling demand after a confirmation is temporary, the price drop should be transient and eventually reversed. Moreover, as the “delay effect” keeps withholding selling demand in this new block cycle, it continuously influences the buy-sell mismatch and results in positive returns beyond the reversal until a new

⁸In unreported results, we find similar confirmation effects among altcoins, such as Bitcoin Cash and Litecoin, which have relatively long wait times designed for security considerations.

⁹There is a complication in reality that could influence how this effect plays out. Specifically, although Binance officially requires at least one block confirmation for a depositing Bitcoin to an exchange wallet, it could temporarily increase depending on the market conditions. No reliable data allow us to directly observe the actual number of confirmations required for each flow. In our reported tests, we assume that one confirmation is required as this should hold the vast majority of the time. In unreported results, we find that wait times of previous blocks have little impact on the confirmation effect of the current block.

confirmation occurs.

We test the first two implications by examining the relationship between the preceding wait time and the post-confirmation return. we identify block confirmations that are preceded by long wait times or overlapped. We define the former as those with a wait time of 15 minutes or more. This number is five minutes more than the target wait time of ten minutes and is roughly the third quartile of the wait time distribution in our sample. We define a confirmation as overlapped if it occurs within one minute of a preceding confirmation. We estimate a regression framework specified as follows:

$$Y_t = \alpha + \beta_1(\text{Post}_t \times \text{Long_wait}_t) + \beta_2(\text{Post}_t \times \text{Overlapped}_t) + \beta_3\text{Post}_t + \beta_4\text{Long_wait}_t + \beta_5\text{Overlapped}_t + \delta_{Reset} + \zeta_{DOW} + \eta_{HOD} + \epsilon_t. \quad (3)$$

Here, Y_t is either the minute-level market order imbalance or return. Post_t is an indicator that equals one if the minute is within five minutes of the most recent confirmation, and zero otherwise. Long_wait is equal to one for confirmations with a preceding wait time of at least 15 minutes, and zero otherwise. Overlapped is equal to one for confirmations within one minute of a preceding one. The coefficients of interest are on the interaction terms $\text{Post} \times \text{Long_wait}$ and $\text{Post} \times \text{Overlapped}$. Our first two implications above predict negative and significant β_1 and β_2 for both the order imbalance and the return regressions.

Table 4 presents the results. We observe strong support for the first two implications. The coefficient on $\text{Post} \times \text{Long_wait}$ is negative and statistically significant across all specifications. In the order imbalance regression in column (1), the coefficient is -0.598 with a highly significant t-statistic of -5.32. Compared to the coefficient on Post , this number indicates that long-awaited block confirmations experience 34% more negative price pressure in the post-confirmation window. The return regressions in columns (2) and (3) suggest that long-awaited block confirmations experience about 58% more negative returns after the confirmation. Overlapped confirmations also experience more negative post-confirmation order imbalance and returns although the effects are both smaller and statistically weaker than among the long-awaited ones. The coefficients on $\text{Post} \times \text{Overlapped}$ suggest a 14% more negative order imbalance and a 33% to 38% more negative return. Overall, these results support the argument that the “delay”

channel influence the confirmation effect through varying inter-block duration.

For the third implication, we examine the continuous relationship between returns and the time elapsed since the last confirmation by estimating the following model:

$$Y_t = \alpha + \beta_1 \log(\text{MinutesSinceConfirmation}_t) + \delta_{Reset} + \zeta_{DOW} + \eta_{HOD} + \epsilon_t. \quad (4)$$

Y_t is either the order imbalance or the return. $\log(\text{MinutesSinceConfirmation}_t)$ is the natural log of the number of minutes since the last confirmation. We predict that β_1 is positive and significant.

In an alternative specification, we also break down the entire period between two blocks into five-minute time bins. We include time bin indicators, each of which equals one if the minute falls into this bin, and zero otherwise. Specifically, this alternative new model is as follows:

$$Y_t = \alpha + \sum_k \beta_k \text{TimeBin}_{k,t} + \delta_{Reset} + \zeta_{DOW} + \eta_{HOD} + \epsilon_t. \quad (5)$$

Here, we include five time bins, [1, 5], [6, 10], [11, 15], [16, 20], and [>21]. For example, [1, 5] is equal to one for the five minutes immediately after the previous confirmation and zero otherwise. We leave the confirmation minute as the benchmark, which is not greatly affected by the confirmation effect according to our results so far. Note that, in this specification, not every block has observations in every time bin.

Panel A of Table 5 shows the estimation of Equation (4). The coefficients on the logarithm of minutes since the last confirmation are positive and highly significant for both returns and order imbalance. After a confirmation event, both the order flow and the returns increase with the time passed. Panel B provides a more granular view by estimating Equation (5). Both the order imbalance and the returns tend to be more negative in the first five minutes, confirming our results in the previous sections. Moreover, supporting the reversal, both the order imbalance and the returns turn more positive than in the confirmation minute after the first five minutes. The initial price drop seems to fully reverse within the next five to fifteen minutes of the [1, 5] bin. Interesting, the price increase persists when the delay of the next confirmation is sufficiently long, resulting positive returns beyond the magnitude of the initial price drop. This result clearly demonstrates how the “delay effect” can cause buy-sell mismatch that drives both

positive and negative returns in different periods.

4.5 The “Delay” channel: Evidence from Exchange Flows

In the previous section, we show that a longer wait time before a block confirmation results in stronger selling pressure, likely due to more Bitcoin inflows to the exchange being accumulated to the upcoming block. We complement the wait-time analyses directly using estimated exchange flows at the block level. Recent studies, including Griffin and Shams (2020) and Makarov and Schoar (2025) use clustering algorithms, supplemented with manual searching, to identify cryptocurrency wallets that likely belong to certain exchanges and study inter-exchange coin flows. This technique is also capitalized by commercial platforms of blockchain analytics. We obtain the block-level inflows and outflows to Binance from CryptoQuant, one of the leading crypto analytic platforms. These variables demonstrate large variations across blocks as shown in Table 1. As such, they can help facilitate a heterogeneity test across blocks as a proxy of the block-associated selling demand.

Following our hypothesis, we expect blocks with large inflows to predict stronger post-confirmation selling pressure and more negative returns. We investigate this flow-related implication using the following model:

$$Y_t = \alpha + \beta_1(\text{Post}_t \times \text{Large_inflow}_t) + \beta_2(\text{Post}_t \times \text{Large_outflow}_t) + \beta_3\text{Post}_t + \beta_4\text{Large_inflow}_t + \beta_5\text{Large_outflow}_t + \beta_6\text{Log_wait}_t + \delta_{\text{Reset}} + \zeta_{\text{DOW}} + \eta_{\text{HOD}} + \epsilon_t. \quad (6)$$

Similar to Equation (3), Y_t is either the minute-level market order imbalance or return. Post_t is an indicator that equals one if the minute is within five minutes of the most recent confirmation, and zero otherwise. Large_inflow (Large_outflow) is equal to one for block confirmations with inflows (outflows) that are higher than the 80 percentile of the previous reset cycle, and zero otherwise. We control the natural log of the wait time to focus on the flow effect beyond Table 4. The coefficients of interest are on the interaction terms. We predict negative and significant β_1 for all specifications while outflows should have no impacts.

The results are reported in Table 6. Consistent with our hypothesis, the coefficient on $\text{Post} \times \text{Large_inflow}$ is negative and statistically significant across all specifications. In column

(1), the coefficient is -0.306 with a significant t-statistic of -2.47. In the return regressions in columns (2) and (3), this coefficient is -0.185 and -0.189, respectively, both highly statistically significant. The magnitudes suggest that blocks with large inflows more than double the negative returns in the post-confirmation window. Large outflows have no impact on the post-confirmation returns. While exchange flows may not be as exogenous as wait time in the previous section, this result complements our analyses on wait time and highlights the interplay between the block confirmations and exchange flows on the return dynamics.

5 Economic Magnitude and Tradable Strategies

The results so far suggest that the confirmation effect is statistically significant and economically large at the minute level. Several questions still need to be answered to help understand the prevalence and market implications of this effect: How did this effect evolve over time? Is the lack of arbitrage due to transaction costs? How profitable are the tradable strategies that exploit this effect?

We provide answers to these questions in two steps. In the first step, we divide all minutes in our sample into the post-confirmation periods, defined as the five minutes immediately after the most recent confirmation, and other periods. We calculate the before-cost cumulative returns of investing exclusively in each of these periods. For actual tradable strategies, the core logic is to short in the post-confirmation periods and long in other periods. In the second step, we take into account the transaction costs and calculate the after-cost cumulative returns of the “Long” and “Short” strategies.

5.1 Magnitude and Persistence of Before-Cost Strategy Returns

Figure 7 plots the log cumulative returns of these strategies against a passive buy-and-hold approach across our sample period. Both the post-confirmation and other periods show stark divergences from the buy-and-hold returns. Investing only in the post-confirmation periods yields a log cumulative return of -17.585 from the beginning of 2018 to the end of 2024. In sharp contrast, investing only in other periods yields a log cumulative return of 19.505 in the same sample. It is also worth noting that these patterns of the log cumulative returns are linear

and largely monotonic, suggesting that these patterns are evenly distributed across our sample years.

Table 7 Panel A translates these patterns to daily before-cost strategy returns. The “Long” and “Short” strategies involves taking long positions outside of post-confirmation periods and taking short positions during post-confirmation periods, respectively. An actual short position may involve leverage and margin requirements, complicating the return calculation. Here, we take a simple view and use the opposite of the long-position return. The “Long + Short” strategy return is calculated by compounding the two strategies as they are active in complementary time periods by design. Across our sample years, the “Long” and “Short” strategies generate daily returns of 0.827% and 0.672%, respectively, both statistically significant at the 1% level. As a result, the combined strategy yields a daily return of 1.515%, also highly significant.

It is important to point out that, although Bitcoin returns are highly volatile, these three strategies are associated with exceptionally high Sharpe ratios of 5.108, 5.172, and 6.828, respectively. To put these numbers in perspective, the daily return of the buy-and-hold strategy is only marginally significant at 0.139% with a Sharpe ratio of 0.690.

The average returns by calendar year reveal a highly persistent confirmation effect. Both the “Long” and “Short” strategies yield highly significant returns in all years across our sample. The combined strategy yields average daily returns ranging from 0.955% (2019) to 2.325% (2021). The Sharpe ratios of the combined strategy range from 4.356 (2019) to 11.266 (2026). Note that two reward-halving events occurred in our sample on May 11th, 2020 and April 19th, 2024, respectively. The average daily returns across the years show no sign of diminishing over time, particularly around the years of these halving events. This result further undermines the potential role of the “reward effect.”

5.2 Transaction Costs and After-Cost Returns

Next, we take into account transaction costs when estimating these strategy returns. The Bitcoin market has been known for its high liquidity and small bid-ask spreads in recent years. As the largest exchange for Bitcoin trading, Binance offers particularly small bid-ask spreads and zero

trading fees on a variety of trading pairs.¹⁰ However, in the early part of our sample, the higher transaction costs could erode the profits of the strategies given their high-frequency nature.

For the “Long” strategy, we take a simple and conservative approach of buying at the ask price and selling at the bid price. The “Short” strategy, however, involves two types of transaction costs, namely the bid-ask spread and the borrowing cost. To incorporate the former, the borrowed positions are sold at the bid price and bought back at the ask price. In addition, we take into account the cost of Bitcoin borrowing by discounting the daily return with the daily borrowing rate on Binance. As Binance API only allows us to track back six months on the borrowing rate information, we utilize ChatGPT to retrieve the borrowing cost estimates for each calendar year in our sample using all available information online. We validate these estimates with a history of online sources related to borrowing rate changes and promotions before applying them as part of the transaction costs.¹¹ Finally, the return of a combined strategy (“Long + Short”) is calculated by compounding the returns of the “Long” and “Short” strategies.

The bid-ask spread plays a key role in the after-cost returns. Therefore, we first plot the average bid-ask spread by calendar year in Figure 8. Each year, the average bid-ask spread is calculated from all the minute-level bid-ask spreads. In 2018, the first year of our sample, the average bid-ask spread is large at 3.484 bps. But this number rapidly drops to 1.467 bps in 2019 and 0.393 in 2020. After that, the spread drops slowly but consistently to as low as 0.042 in 2024, the last year of our sample. Given these bid-ask spreads, the strategies above may be able to survive the transaction costs, especially in the later years where the spreads are particularly small.

Panel B of Table 7 reports the average after-cost daily returns. Across our sample years, the “Long” and “Short” strategies generate a daily returns of 0.144% (t-stat=2.18) and 0.037% (t-stat=0.68), respectively. The combined strategy yields an average daily return of 0.204 (t-

¹⁰Binance completely eliminated trading fees in a period from 2022 to 2023. In other periods, Binance offers zero-trading-fee promotions on a subset of trading pairs, which varies over time. We exclude trading fee in our analyses here.

¹¹ChatGPT provides a range estimate for each year in our sample. From 2018 to 2024, the midpoints of the ranges are 11.5%, 9%, 7.5%, 5.5%, 4%, 2.3%, and 2%, all quoted as annual rates. These numbers are consistent with evidence from news. For example, a news article from September 2020 quotes the Bitcoin borrowing rate at 10.03%. CoinMarketCap estimated the Bitcoin borrowing rate at 5% in 2021. In June of 2023, Binance reduced the borrowing rate on Bitcoin from 1.91% to 1% in a promotion. Binance started crypto lending in August 2019 and, therefore, related information before that is scarce. The borrowing interest is charged on an hourly basis and calculated from the annual borrowing rate. Because the borrowing rate is trivial compared to the daily strategy returns, variations in the estimates have little impact on the after-cost returns.

stat=2.16). Comparing these numbers to those in Panel A, the transaction costs erode a large part of the daily return on average. This is primarily due to the interplay between the high-frequency nature of this strategy and the large bid-ask spreads in the first two years of our sample as shown in Figure 8. Both the “Long” and “Short” strategies are associated with large negative returns in 2018 and 2019. Starting 2020, however, both strategies generate positive returns. The combined strategy yields average daily returns ranging from 0.749% (2020) to 1.691% (2024), all highly significant. The corresponding Sharpe ratios range from 3.684 (2020) to 10.322 (2023).

Turning to the after-cost Sharpe ratios, the combined strategy generates a Sharpe ratio of 3.413 across all the years in spite of the negative returns in 2018 and 2019. Starting 2020, This strategy generates Sharpe ratios ranging from 4.892 (2020) to 15.935 (2024). These numbers depict an exceptional risk-return tradeoff of these strategies exploiting the Bitcoin block confirmation effect, ruling out arbitrage risk as an explanation. Considered together with the large positive after-cost returns, these results suggest that the existence of the confirmation effect is not explained by costly arbitrage. Instead, it points towards lack of sophisticated investors as a potential explanation for this market inefficiency.

The log cumulative after-cost returns of these strategies are plotted in Figure 9. We exclude 2018 and 2019 from this plot because 1) the bid-ask spreads are too large in these years and 2) Binance did not offer crypto lending services until August 2019. Both the “Long” and “Short” strategies suffered in the first half of 2020 because the bid-ask spread was still large. After this period, both strategies start to generate positive returns persistently, with the “Long” strategy reaching 13.207 and the “Short” strategy reaching 10.437 by the end of our sample. That is, one dollar invested in the “Long” (“Short”) strategy at the beginning of 2020 appreciates to 544,161 dollars (34,098 dollars) by the end of 2024. The combined strategy has a log cumulative return of 23.643, which translates to unachievable dollar gains in reality due to market frictions. Overall, these plots further highlight the economic meaning of this confirmation effect.

6 Confirmation Effect Spillover and Crypto-Market Comovement

Our last set of results link the block confirmations of Bitcoin to the price dynamics of other popular cryptocurrencies. The previous studies, such as Hu et al. (2018), Liu and Tsyvinski (2021), and Liu et al. (2022), document price comovement of cryptocurrencies in the secondary market. In particular, Liu et al. (2022) show that a cryptocurrency market factor is crucial in explaining cryptocurrency portfolio returns. Since Bitcoin is the largest constituent of the crypto market index, its price movement could influence returns of other cryptocurrencies through the systemic channel. However, the confirmation effect documented above is clearly transient, unrelated to Bitcoin fundamentals, and theoretically has no market-wide implications. In an efficient market, investors should be able to filter out these noises and thus prevent a spillover effect across cryptocurrencies.

We investigate whether such a spillover effect exists on the four most popular altcoins in terms of market capitalization, including Ethereum (ETH), BNB, XRP, and Solana (SOL). Figure 10 plots the order imbalance and returns of these altcoins around the time of Bitcoin block confirmations. A discernible pattern of negative returns and order imbalances emerges for these altcoins, closely mirroring the timing of the confirmation effect on Bitcoin itself, although with smaller magnitudes.

This result is crucial to our understanding of how prices behave in the crypto-market. A market factor should explain the common component of price movements through market-wide effects. However, based on a clearly identified event study framework, we find that transient and non-fundamental shocks in Bitcoin trading and returns also propagate to other cryptocurrencies, likely through investors' inattentive cross-cryptocurrency arbitrage. This raises the question as to how much comovement in the cryptocurrency market is driven by noisy arbitrage activities. In other words, how much are common risks in this market attributable to transient and non-fundamental price movement?

7 Conclusion

In this paper, we test our hypothesis that block confirmations create swiftly changing mismatch in trading demand in the secondary market through the “delay” channel. Using high-frequency Bitcoin data, we show that each confirmation triggers a burst of sell-side trading activity and a transient yet highly significant price decline. This confirmation effect is positively associated with the pre-confirmation wait time. After the initial price drop, however, the price gradually recovers and then exceeds its pre-confirmation level in absence of new block confirmations. Additional results based on exchange flows further support the negative price pressure from the delayed exchange inflows.

The economic magnitude of this effect is substantial. Simple trading strategies designed to exploit this predictability generate persistently high returns in our sample period. These strategies continue to generate large returns and exceptional Sharpe ratios even after accounting for transaction costs, ruling out costs of arbitrage as an explanation.

Beyond Bitcoin itself, we find that these confirmation-induced shocks spill over to other major cryptocurrencies, producing short-term price comovement. This spillover effect implies that transient and non-fundamental price movements in the largest cryptocurrency can cause market-wide comovement, urging us to rethink about the content of the common risk factors in this market.

Our results contribute to the growing literature on the microstructure of decentralized markets. By identifying a predictable and economically significant source of price pressure, we provide evidence that blockchain design choices—such as block-generation frequency—play a first-order role in shaping the behavior of asset prices and overall market efficiency.

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Appendix: Institutional Background of Bitcoin Mining

This appendix outlines the core institutional mechanisms of Bitcoin’s Proof-of-Work consensus as defined in the original protocol . These include the mining process, difficulty adjustment, and block reward schedule, which together ensure decentralized security, predictable monetary growth, and a near-random block creation process in the short run.

Proof-of-Work Mining

Blockchain is updated in a decentralized manner. In Proof-of-Work (PoW) systems, all participating miners have a probability—proportional to their computational power—of solving the cryptographic puzzle and thereby gaining the right to append a new block to the ledger.

Proof-of-Work mining involves repeatedly hashing the block header until a hash value is found that satisfies the network’s difficulty target—specifically, one that is below a predefined threshold. This process includes verifying transactions and assembling them into a candidate block. Crucially, miners must find a nonce (a natural number) such that the hash of the block header meets this criterion. Since cryptographic hash functions (e.g., SHA-256) are one-way functions, it is computationally infeasible to reverse-engineer an input from a desired output. Thus, the only feasible method is trial and error—hashing vast numbers of candidate inputs. This makes the process computationally intensive and inherently unpredictable.

The first miner to find a valid solution broadcasts it to the network for validation. Upon consensus, the block is added to the blockchain, and the miner receives a block reward. This reward mechanism incentivizes participation and secures the network against attacks.¹²

Over time, mining has become increasingly competitive. As individual success probabilities have diminished due to rising network difficulty and specialized hardware (ASICs), miners have formed *mining pools* to combine their computational resources and share rewards proportionally. Today, the majority of Bitcoin mining occurs via pool mining rather than solo mining.

¹²In Bitcoin, the block reward halves approximately every 210,000 blocks (roughly every four years), ensuring a predictable and diminishing supply of new coins.

Minting Events

A *minting event* occurs when a miner successfully finds a valid block, triggering the creation of a new block and the issuance (“minting”) of new coins as a block reward. In Bitcoin, these events occur on average every 10 minutes, though the actual inter-arrival times vary stochastically.

Critically, the timing of minting events constitutes a near-random natural experiment. The data-generating process approximates random assignment due to the proof-of-work mechanism and the irreversible, deterministic nature of cryptographic hash functions.¹³

Because miners essentially perform brute-force searches over possible nonces, and because each attempt is independent and uniformly likely to succeed (given fixed difficulty), the time until a solution is found is unpredictable. Even if no solution has been found for several minutes, the expected waiting time for the next success remains unchanged. This *memoryless property* distinguishes blockchain block arrivals from most recurring economic events (?).

Formally, let X denote the random variable representing the time (in minutes) between two consecutive minting events. Under constant network conditions, X follows an exponential distribution with rate parameter $\lambda = 1/10$, corresponding to a Poisson process with an average arrival rate of one block per 10 minutes. The survival function of X satisfies:

$$P(X > m + t \mid X > t) = \frac{P(X > m + t)}{P(X > t)} = \frac{e^{-\lambda(m+t)}}{e^{-\lambda t}} = e^{-\lambda m} = P(X > m), \quad (7)$$

where $m, t \geq 0$. This confirms the memoryless property: the probability of waiting an additional m minutes for the next block does not depend on how long one has already waited.

Consequently, regardless of how much time has passed since the last block, the expected time until the next block remains 10 minutes. This implies that neither miners nor investors can gain an advantage by “timing” the network based on elapsed time since the last block.

While block time depends on network difficulty and total hash rate, short-term fluctuations mean that actual block intervals are variable. Nevertheless, under stable conditions, the long-run average block time converges to 10 minutes. The exponential distribution of inter-

¹³Bitcoin employs the Secure Hash Algorithm 256 (SHA-256), a widely used cryptographic hash function developed by the NSA. This paper focuses on cryptocurrencies employing similar PoW mechanisms; proof-of-stake systems generally lack this exogenous variation in block timing.

arrival times ensures that predictions about the next block should treat the process as memoryless—consistent with a Poisson arrival process.

This stochasticity in block arrivals has direct economic consequences, introducing significant income volatility for miners. For an individual (solo) miner, revenue follows a high-variance "jackpot" model, where finding a block is a rare event. This uncertainty is the primary economic incentive for the formation of mining pools, which aggregate the computational power of many participants to smooth revenue streams, distributing rewards proportionally to contributed work (or "shares") rather than discrete block-finds. Moreover, the protocol imposes an additional temporal constraint on these newly minted assets. The coinbase transaction—the transaction containing the block reward—is subject to a "coinbase maturity" rule. These new coins are unspendable by the miner until 100 additional blocks have been confirmed (a period of roughly 16-17 hours), a mechanism designed to ensure chain stability and prevent miners from spending rewards from blocks that are subsequently orphaned.

Difficulty Adjustment Mechanism

To maintain an average block time of 10 minutes despite fluctuations in computational power, Bitcoin adjusts the mining difficulty every 2,016 blocks (approximately every two weeks). The new difficulty is proportional to the time taken to mine the previous 2,016 blocks:

$$\text{Difficulty}_{\text{new}} = \text{Difficulty}_{\text{old}} \times \frac{T_{\text{actual}}}{2,016 \times 600}, \quad (8)$$

where T_{actual} is the actual time (in seconds) to mine the last 2,016 blocks, and $2,016 \times 600 = 1,209,600$ seconds is the target duration (two weeks at 10-minute intervals) .

This adjustment ensures that if blocks are mined too quickly, the difficulty increases, and vice versa. Between adjustments, the difficulty remains constant, so the block generation process remains homogeneous and memoryless within each epoch.

Block Reward and Halving Schedule

The block reward consists of newly issued bitcoins (the *coinbase*) and transaction fees. Initially set at 50 BTC per block, the coinbase reward is programmed to *halve* every 210,000 blocks.

This halving process will continue until the total supply approaches the hard cap of 21 million bitcoins, expected around the year 2140 .

While the halving is deterministic and publicly scheduled, the timing of individual block creations remains random due to the PoW mechanism. This creates a unique institutional structure: long-term monetary policy is fully predictable, but short-term block arrivals are effectively random.

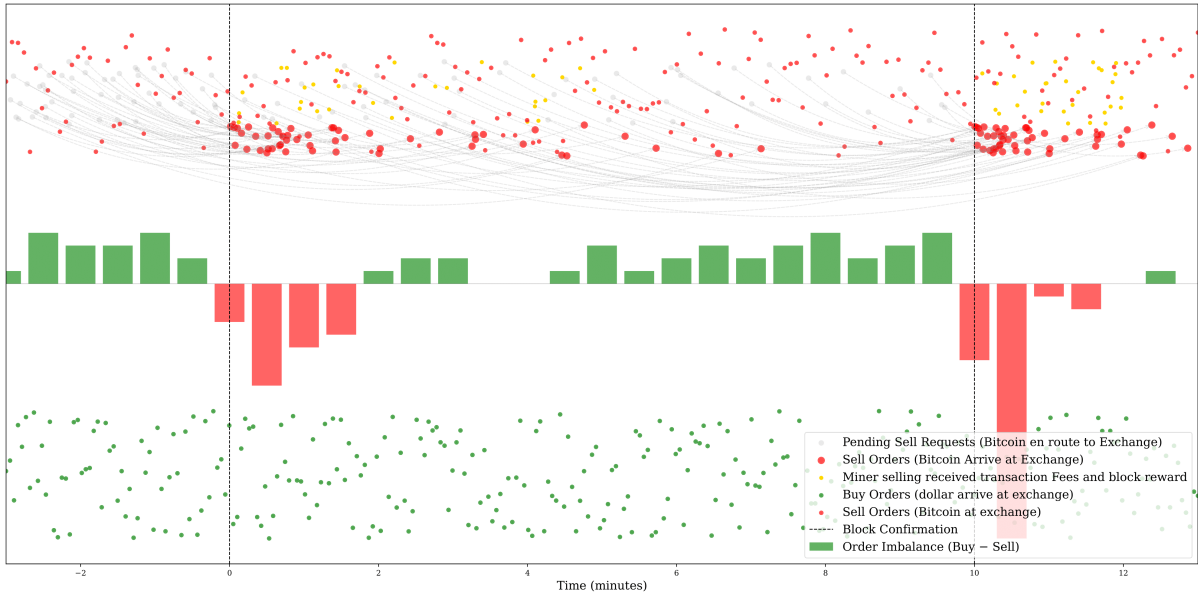


Figure 1: Illustration of the “reward” and “delay” channels

This figure illustrates how block confirmations influence Bitcoin trading through the “reward” and “delay” channels. The red and green dots represent the randomly arrived selling and buying demands, respectively. The gray dots represent the demands to sell Bitcoins in an off-exchange wallet, which can only be executed after a block confirmation that validates their arrival at the exchange. The yellow dots are the demands from winning miners to sell their winning rewards.

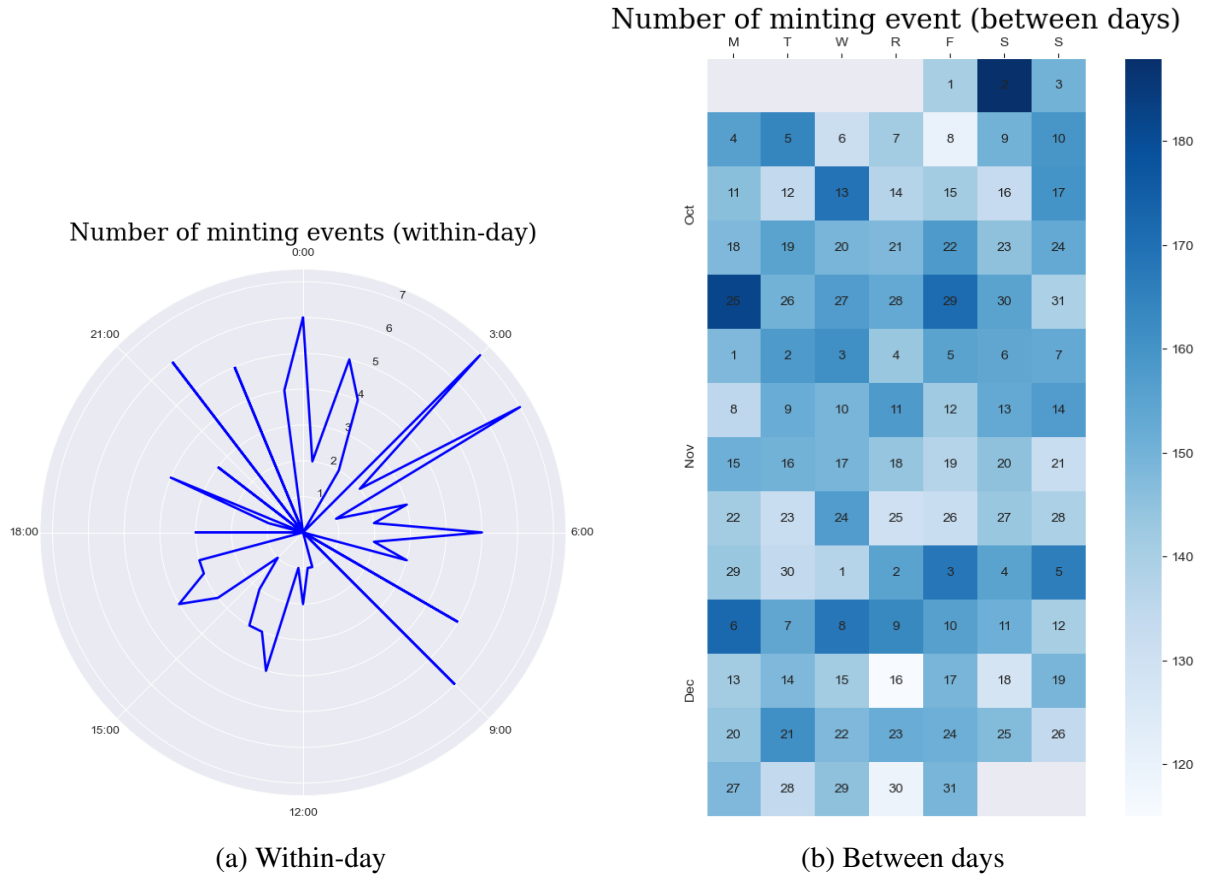


Figure 2: Randomness of Bitcoin block confirmations

This figure demonstrates the distribution of block confirmations in a typical 24-hour window and in a typical three-month period. Panel (a) plots the number of block confirmations in each 30-minute bin of Dec 12, 2021. Panel (b) shows the heat map of the number of block confirmations by calendar day for the three months from October to December 2021. The target number of confirmations per day by design is 144.

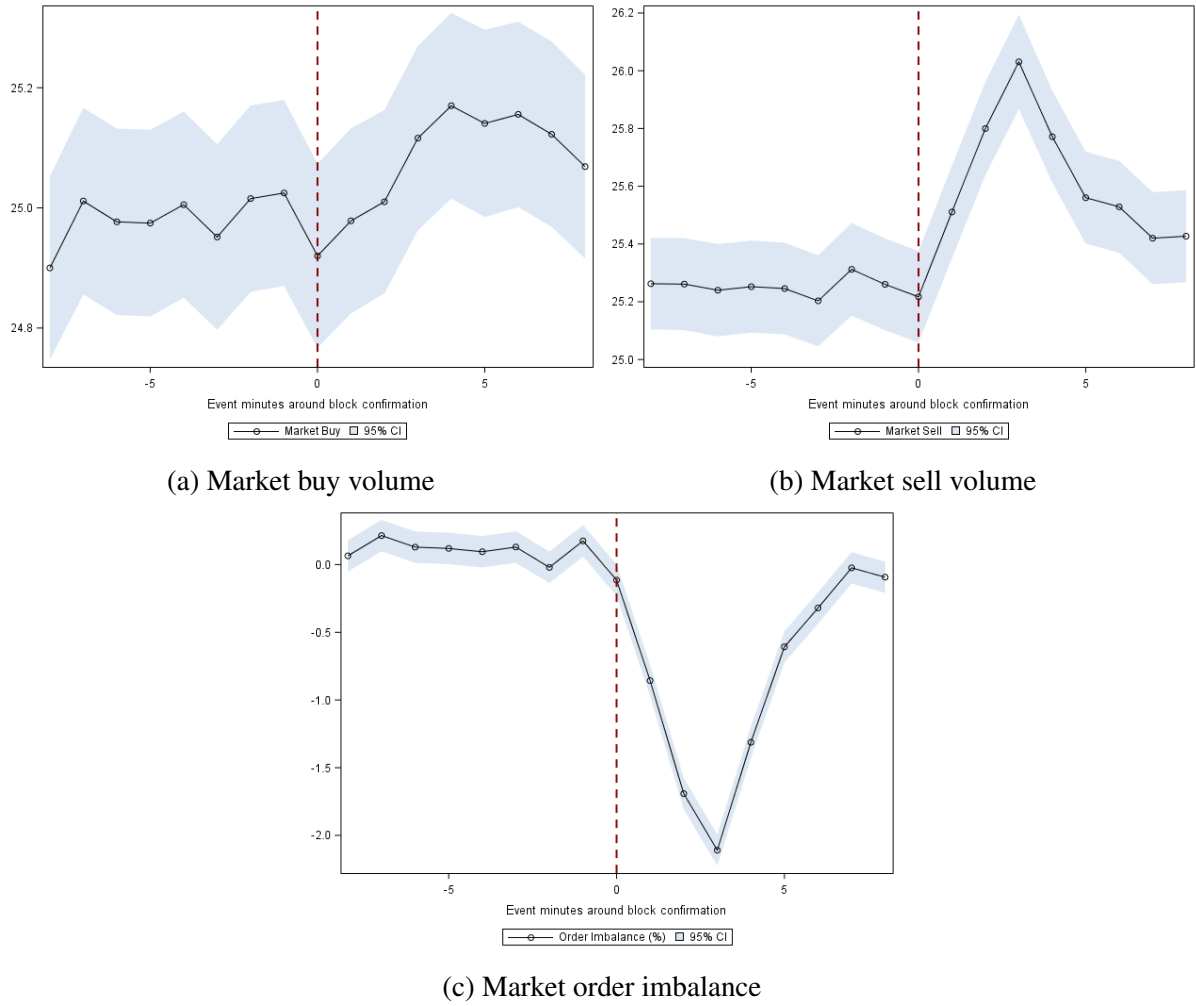
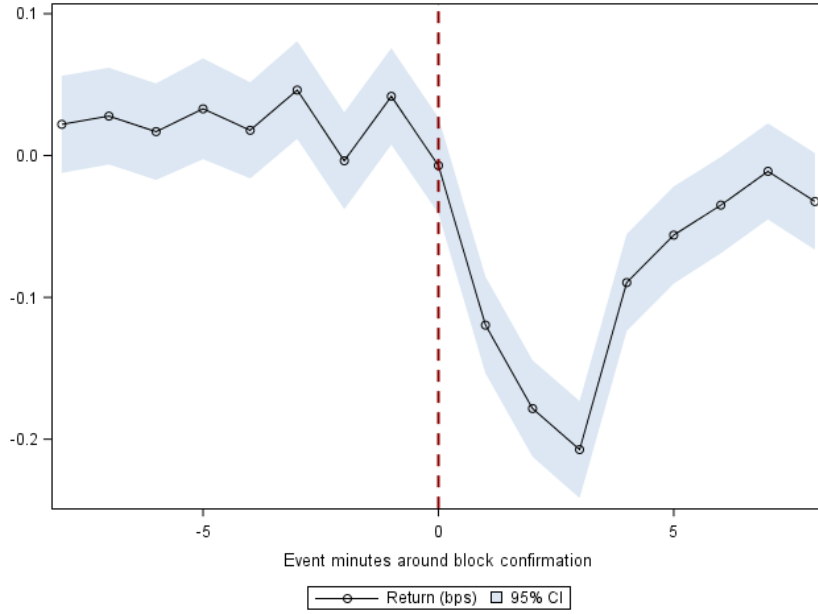
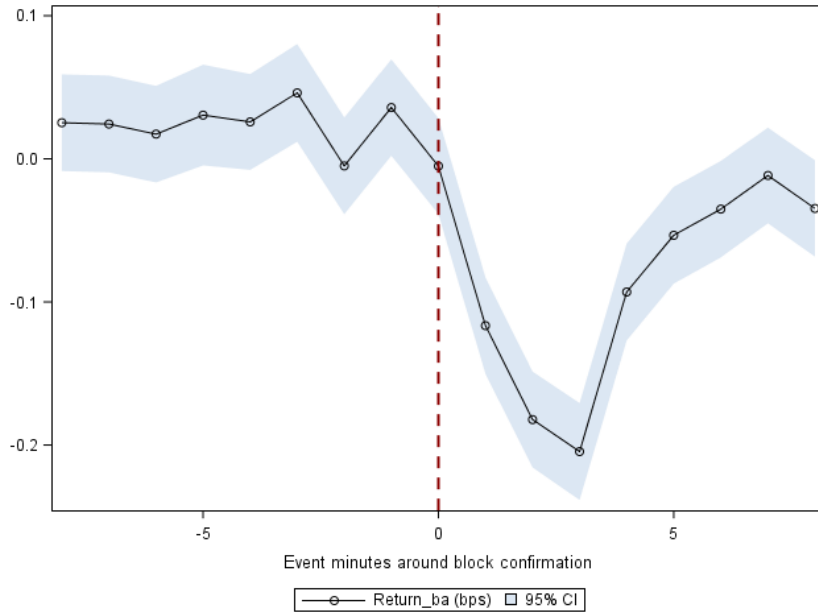


Figure 3: Bitcoin trading around block confirmations

This figure demonstrates trading activities around block confirmations. The sample covers from 2018 to 2024 and the Bitcoin to USDT trading on Binance in the same period. Panels (a) and (c) plot the average minute-level buying and selling volumes using market orders in Bitcoins, respectively, from eight minutes before to eight minutes after block confirmations. Panel (c) plots the average minute-level market order imbalance around block confirmations. The market order imbalance calculated as the difference between the market buy and market sell scaled by the sum of the two.



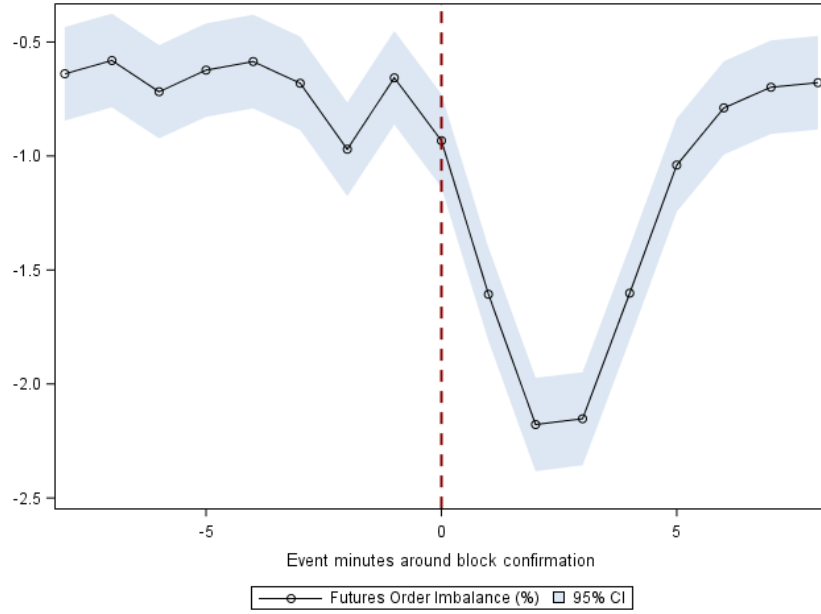
(a) Closing-price return



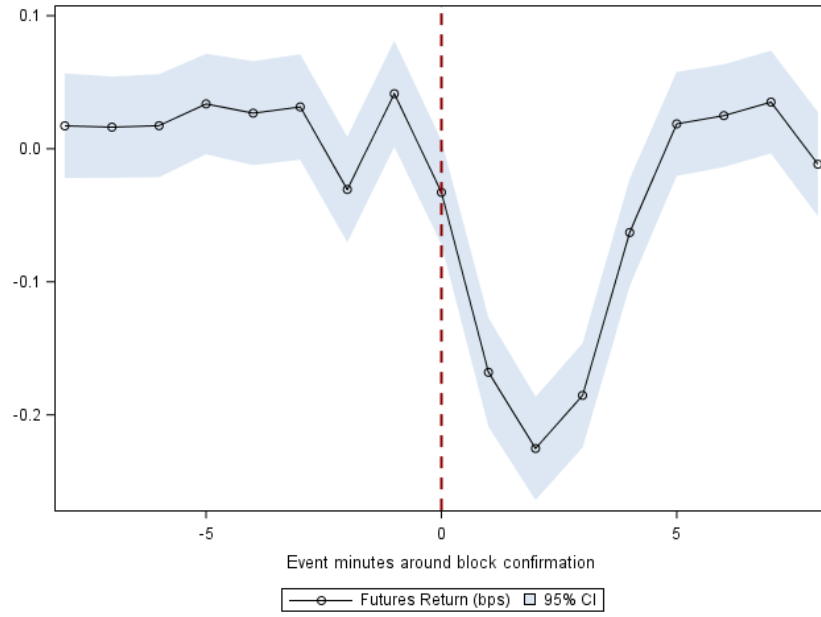
(b) Bid-ask midpoint return

Figure 4: Bitcoin returns around block confirmations

This figure demonstrates minute-level returns around block confirmations. The sample covers from 2018 to 2024 and the Bitcoin to USDT trading on Binance in the same period. Panels (a) and (b) plot the average minute-level returns from eight minutes before to eight minutes after block confirmations using closing-price return and bid-ask midpoint return, respectively. The bid and ask prices are from the last pair of transactions of each minute executed on bid and ask prices, respectively.



(a) Futures order imbalance



(b) Futures return

Figure 5: Bitcoin futures around block confirmation

This figure demonstrates the minute-level order imbalances and returns of Bitcoin futures around block confirmations. The sample covers from 2018 to 2024 and the Bitcoin futures trading on Binance in the same period. Panel (a) plots the average minute-level market order imbalance around block confirmations. The market order imbalance is calculated as the difference between the market buy and market sell scaled by the sum of the two. Panel (b) plots the average minute-level returns around block confirmations using closing prices.

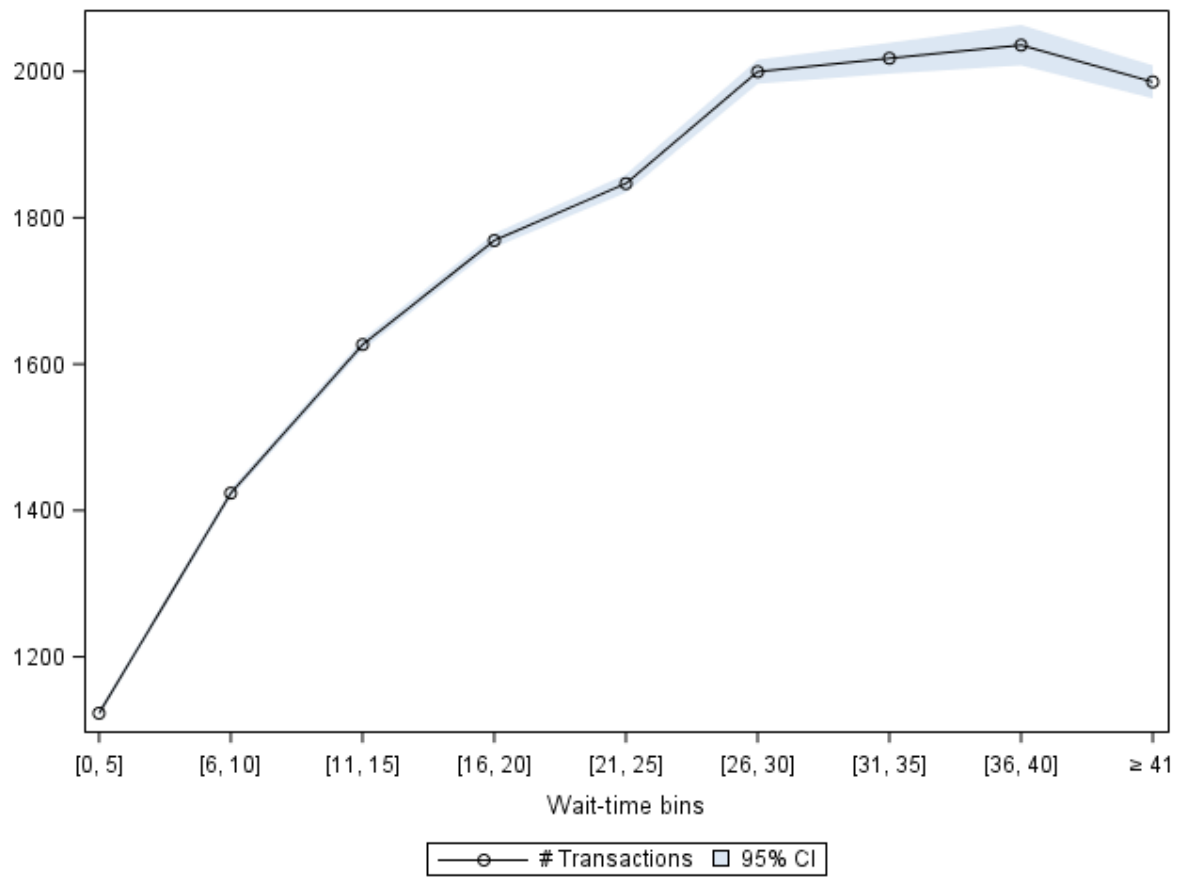


Figure 6: Wait time before confirmation and number of transactions processed

This figure demonstrates the number of transactions processed by blocks with different wait times. The sample covers from 2018 to 2024. Blocks are grouped into different wait-time bins on the horizontal axis.

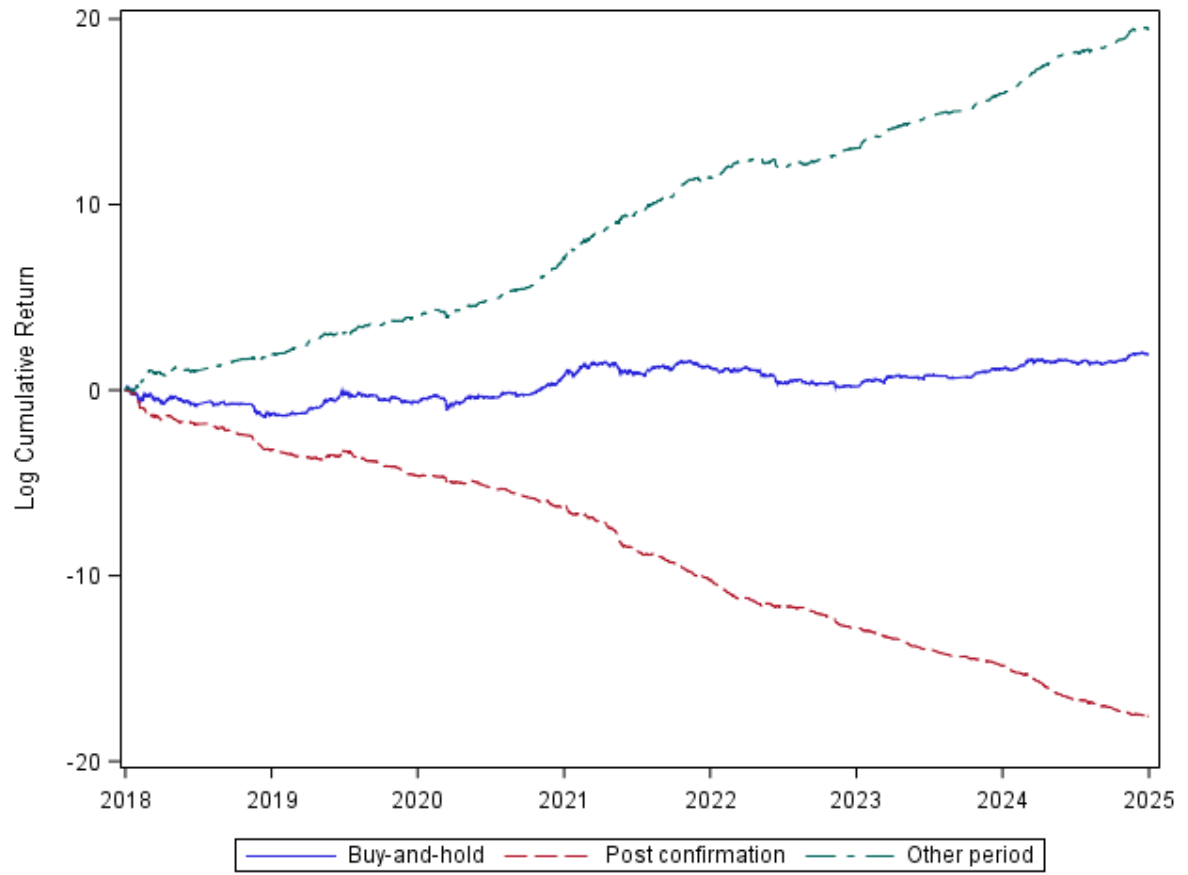


Figure 7: Cumulative return for post-confirmation and other periods

This figure demonstrates the log cumulative returns by investing in only post-confirmation periods or in only other periods. The sample covers from 2018 to 2024 and the Bitcoin to USDT trading on Binance in the same period. The returns are calculated using closing prices. The results based on the bid-ask midpoint returns are similar.

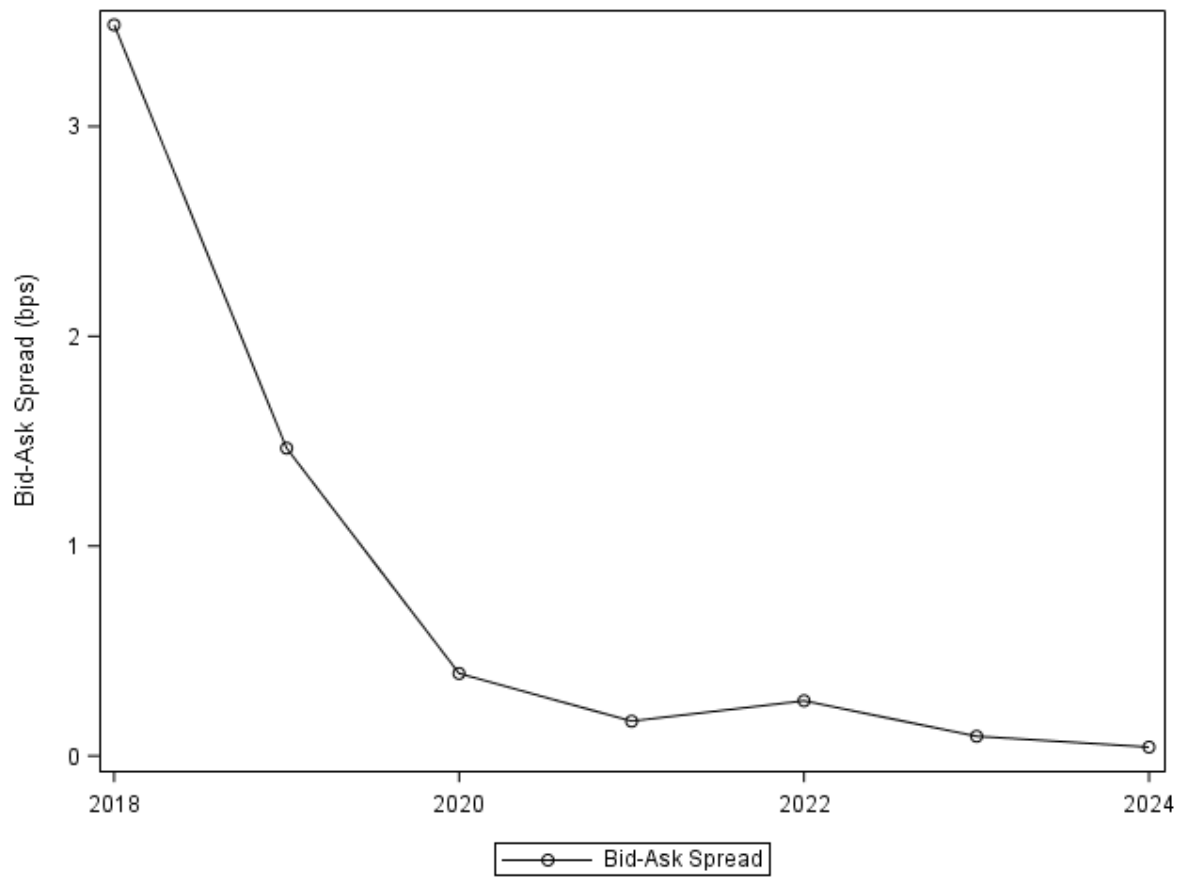


Figure 8: Average bid-ask spread by calendar year

This figure demonstrates how the bid-ask spread evolves over time. The sample covers from 2018 to 2024 and the Bitcoin to USDT trading on Binance in the same period. We plot the average bid-ask spread by calendar year. The bid and ask prices are from the last pair of transactions of each minute executed on bid and ask prices, respectively.

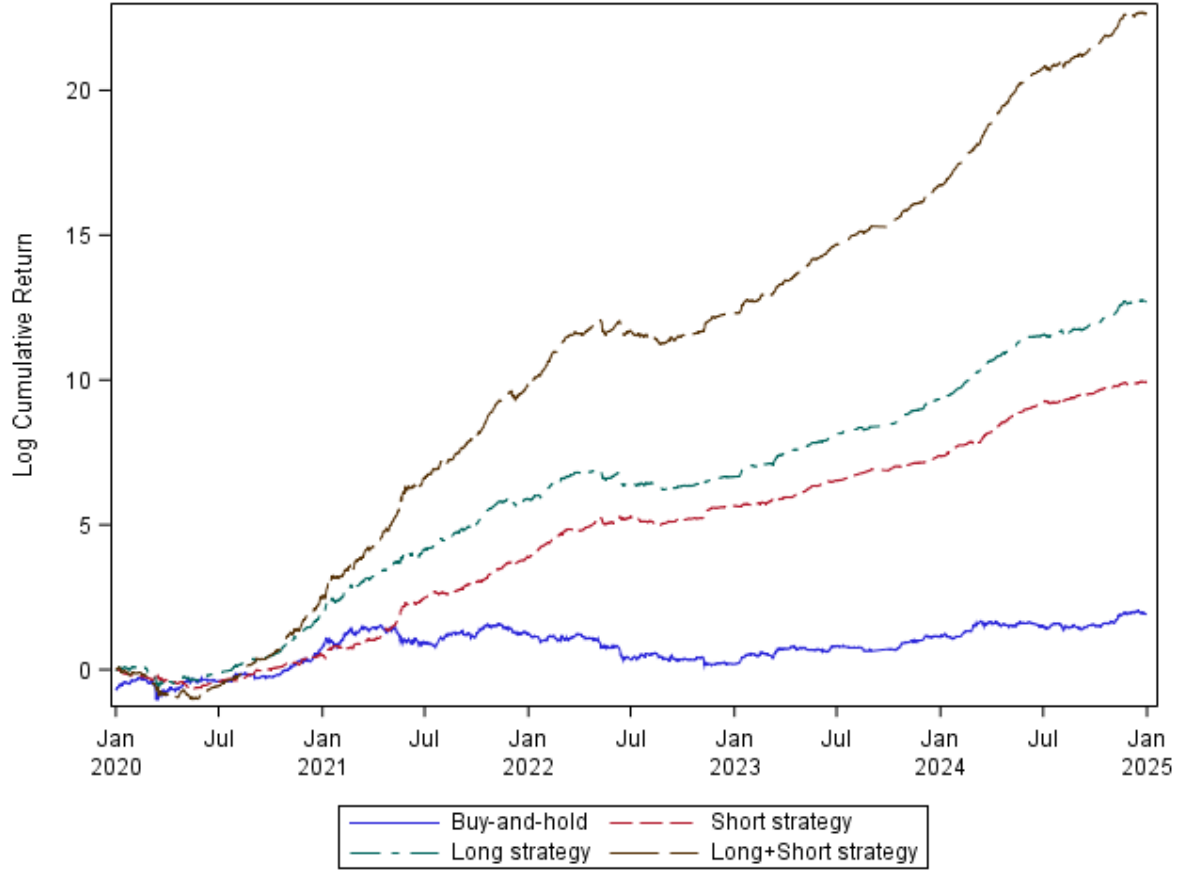
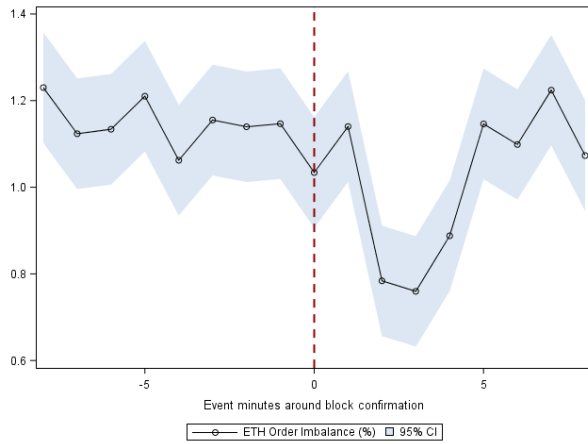
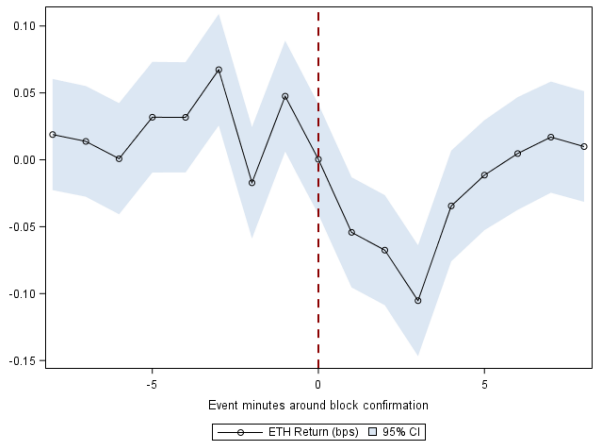


Figure 9: Bitcoin bid-ask spread and after-cost returns

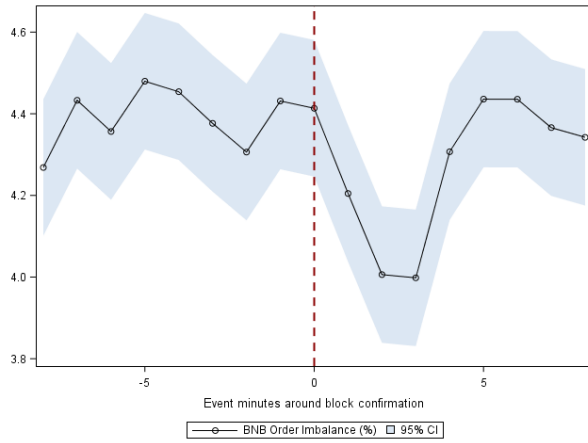
This figure demonstrates the after-cost performance of strategies based on confirmation effect. This figure covers a subsample from 2020 to 2024. We exclude 2018 and 2019 where bid-ask spreads were large due to low liquidity and Binance did not offer crypto lending services. We plot the log cumulative returns of four strategies in our sample period. The buy-and-hold strategy buys Bitcoin at the beginning of our sample period and hold till the end. The long strategy invests only in periods other than the post-confirmation periods, with buying at the ask prices and selling at the bid prices. The short strategy involves borrowing at the margin interest rate and shorting Bitcoin in the post-confirmation periods, with selling at the bid prices and buying back at the ask prices. The last strategy combines the long and short strategies. The post-confirmation periods are defined as the minutes where the most recent block confirmation occurred within five minutes. The bid and ask prices are from the last pair of transactions of each minute executed on bid and ask prices, respectively.



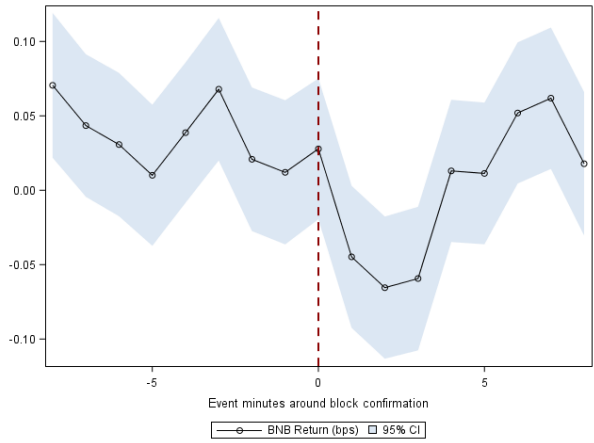
(a) ETH order imbalance



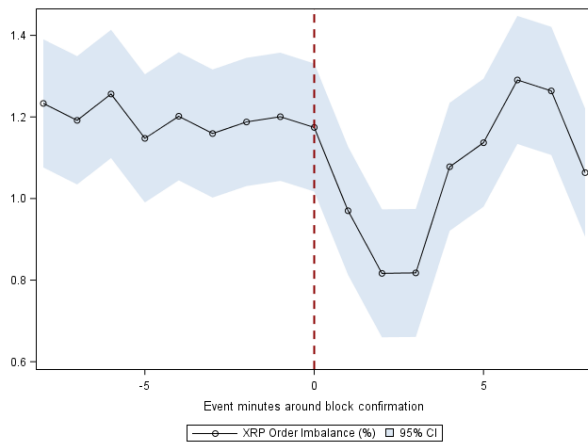
(b) ETH return



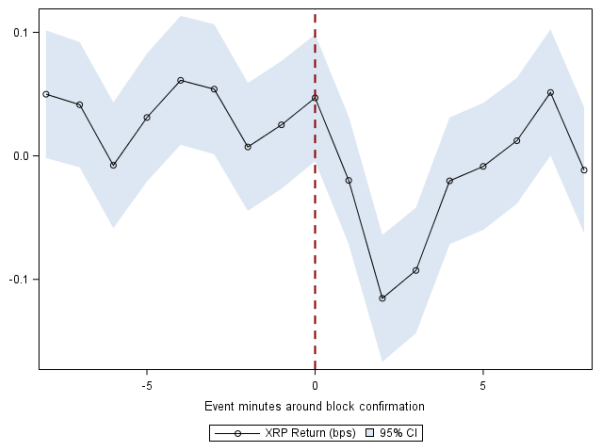
(c) BNB order imbalance



(d) BNB return



(e) XRP order imbalance



(f) XRP return

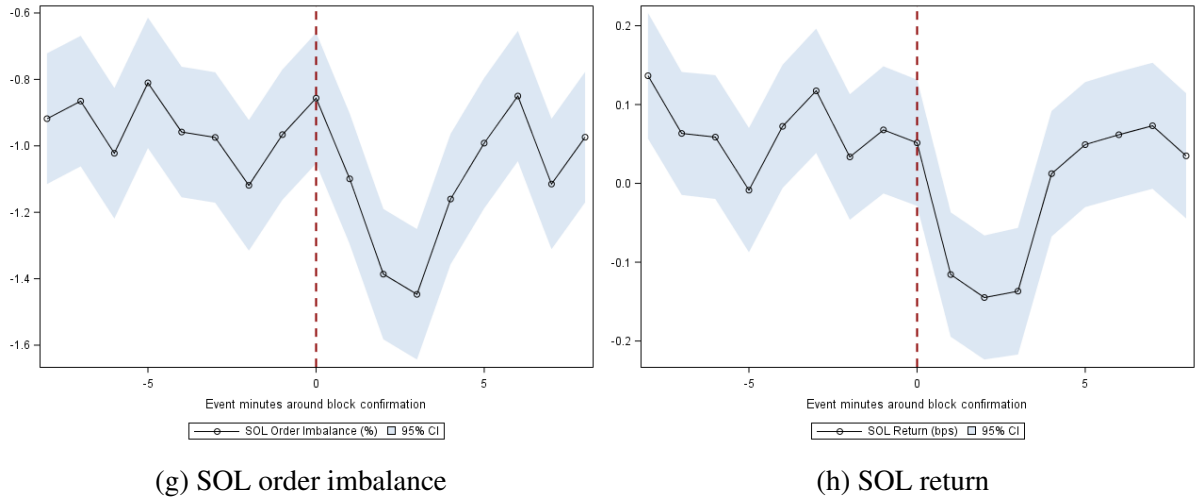


Figure 10: Other cryptocurrencies around Bitcoin block confirmations

This figure demonstrates the minute-level order imbalances and returns of other popular cryptocurrencies around Bitcoin block confirmations. The sample covers from 2018 to 2024 and the price and trading data of Ethereum (ETH), BNB, XRP, and Solana (SOL) on Binance in the same period. Panels (a), (c), (e), (g) plot the average minute-level market order imbalance around Bitcoin block confirmations. The market order imbalance is calculated as the difference between the market buy and market sell scaled by the sum of the two. Panels (b), (d), (f), (h) plot the average minute-level returns around Bitcoin block confirmations using closing prices.

Table 1: Summary statistics

This table reports the summary statistics of the sample at the block level and the minute level. The sample covers from 2018 to 2024 and the Bitcoin to USDT trading on Binance in the same period. For block-level variables, Wait_time is the number of minutes between the previous and current block confirmations. # Transactions is the number of transactions confirmed by the current block. Minted is the number of Bitcoins minted in the current block confirmation. Transaction_fee is the total transaction fees attached to the transactions in the current block. Inflows and outflows to Binance are in number of Bitcoins. For minute-level variables, Return and Return.ba are the minute-by-minute Bitcoin returns based on the minute-level closing price and the bid-ask price midpoint, respectively, in basis points. The bid-ask spread (BA spread) is scaled by the bid-ask price midpoint and is in basis points. The bid and ask prices are from the last pair of transactions of each minute executed on bid and ask prices, respectively. Market buy and Market sell are the market-order buying and selling volumes in number of Bitcoins. Ordimb is the market order imbalance calculated as the difference between the market buy and market sell scaled by the sum of the two.

Block-level variables							
	# Obs	Mean	P10	P25	P50	P75	P90
Wait_time	375163	9.815	1	3	7	14	23
# Transactions	375163	2266	592	1388	2310	2919	3614
Minted	375163	8.071	6.25	6.25	6.25	12.5	12.5
Transaction_fee	375163	0.366	0.032	0.073	0.167	0.394	0.909
Inflow	375163	83.523	1.225	8.268	31.906	86.223	190.964
Outflow	375163	80.843	0.000	1.823	26.914	89.509	206.520
Minute-level variables							
	# Obs	Mean	P10	P25	P50	P75	P90
Return	3586325	0.011	-8.790	-3.534	0.000	3.535	8.808
Return.ba	3585019	0.010	-8.679	-3.462	0.000	3.462	8.695
BA spread	3585073	0.816	0.001	0.002	0.006	0.667	2.395
Market buy	3586325	25.076	2.430	5.131	11.130	25.408	58.142
Market sell	3586325	25.382	2.342	4.971	11.108	25.840	59.025
Ordimb	3586325	0.035	-46.721	-22.817	0.081	23.134	46.819

Table 2: Post-confirmation Bitcoin trading

This table reports the regressions of minute-by-minute Bitcoin trading on block confirmation events. The sample covers from 2018 to 2024 and the Bitcoin to USDT trading on Binance in the same period. The dependent variables include the natural log of market-order buying (LogBuy), the natural log of market-order selling (LogSell), and the market order imbalance in percentage (Ordimb). $Post_m$ is the indicator that the most recent block confirmation was m minutes ago. Reset fixed effects capture observations between difficulty resets, which are 2016 blocks away from each other. DOW and HOD capture the day-of-the-week and hour-of-the-day fixed effects, respectively. Standard errors are clustered at the reset level with the corresponding t-statistics reported in the brackets below. *, **, or *** indicates significance level of 10%, 5%, or 1%, respectively.

VARIABLES	(1) LogBuy	(2) LogSell	(3) Ordimb
Post1	0.002 [1.34]	0.033*** [13.52]	-1.636*** [-13.03]
Post2	0.006*** [3.15]	0.051*** [15.20]	-2.391*** [-12.32]
Post3	0.012*** [6.00]	0.064*** [16.78]	-2.732*** [-13.10]
Post4	0.015*** [8.46]	0.049*** [16.26]	-1.821*** [-12.42]
Post5	0.014*** [7.75]	0.032*** [13.87]	-1.020*** [-8.79]
Constant	2.580*** [4,776.17]	2.566*** [2,637.02]	0.739*** [14.23]
Observations	3,585,011	3,585,011	3,585,011
R-squared	0.456	0.460	0.017
Reset FE	Yes	Yes	Yes
DOW FE	Yes	Yes	Yes
HOD FE	Yes	Yes	Yes

Table 3: Post-confirmation Bitcoin returns

This table reports the regressions of minute-by-minute Bitcoin trading on block confirmation events. The sample covers from 2018 to 2024 and the Bitcoin to USDT trading on Binance in the same period. The dependent variables include the minute return based on closing price and bid-ask price midpoint, both in basis points. The bid and ask prices are from the last pair of transactions of each minute executed on bid and ask prices, respectively. Postm is the indicator that the most recent block confirmation was m minutes ago. Reset fixed effects capture observations between difficulty resets, which are 2016 blocks away from each other. DOW and HOD capture the day-of-the-week and hour-of-the-day fixed effects, respectively. Standard errors are clustered at the reset level with the corresponding t-statistics reported in the brackets below. *, **, or *** indicates significance level of 10%, 5%, or 1%, respectively.

VARIABLES	(1) Return	(2) Return_ba
Post1	-0.206*** [-9.45]	-0.207*** [-9.34]
Post2	-0.255*** [-12.11]	-0.255*** [-12.56]
Post3	-0.274*** [-11.21]	-0.275*** [-11.25]
Post4	-0.147*** [-6.46]	-0.149*** [-6.60]
Post5	-0.099*** [-4.02]	-0.098*** [-4.17]
Constant	0.084*** [16.66]	0.084*** [16.57]
Observations	3,585,011	3,585,011
R-squared	0.000	0.000
Reset FE	Yes	Yes
DOW FE	Yes	Yes
HOD FE	Yes	Yes

Table 4: Pre-confirmation wait time and confirmation effect

This table reports the regressions of minute-by-minute Bitcoin trading and returns on block confirmation events with heterogeneous wait times. The sample covers from 2018 to 2024 and the Bitcoin to USDT trading on Binance in the same period. The dependent variables include the market order imbalance (Ordimb) in percentage and the minute return based on closing price and bid-ask price midpoint in basis points. The bid and ask prices are from the last pair of transactions of each minute executed on bid and ask prices, respectively. Post is an indicator that equals one if the most recent block confirmation occurred within five minutes, and zero otherwise. Long_wait is an indicator that equals one if the wait time before the most recent block confirmation is greater or equal to 15 minutes. Overlapped is an indicator that equals one if the wait time before the most recent block confirmation is less than one minute. Reset fixed effects capture observations between difficulty resets, which are 2016 blocks away from each other. DOW and HOD capture the day-of-the-week and hour-of-the-day fixed effects, respectively. Standard errors are clustered at the reset level with the corresponding t-statistics reported in the brackets below. *, **, or *** indicates significance level of 10%, 5%, or 1%, respectively.

VARIABLES	(1) Ordimb	(2) Return	(3) Return_ba
Post×Long_wait	-0.598*** [-5.32]	-0.099*** [-2.91]	-0.099*** [-2.91]
Post×Overlapped	-0.242* [-1.82]	-0.057* [-1.86]	-0.064** [-2.06]
Post	-1.775*** [-13.33]	-0.171*** [-10.92]	-0.170*** [-10.76]
Long_wait	0.134* [1.76]	0.046** [2.37]	0.046** [2.39]
Overlapped	-0.261*** [-2.84]	0.009 [0.42]	0.013 [0.59]
Constant	0.743*** [14.40]	0.071*** [9.81]	0.071*** [9.55]
Observations	3,585,011	3,585,011	3,585,011
R-squared	0.017	0.000	0.000
Reset FE	Yes	Yes	Yes
DOW FE	Yes	Yes	Yes
HOD FE	Yes	Yes	Yes

Table 5: Time from block confirmation and Bitcoin returns

This table reports the regressions of minute-by-minute Bitcoin trading and returns on time from the previous block confirmation. The sample covers from 2018 to 2024 and the Bitcoin to USDT trading on Binance in the same period. The dependent variables include the market order imbalance (Ordimb) in percentage and the minute return based on closing price and bid-ask price midpoint in basis points. The bid and ask prices are from the last pair of transactions of each minute executed on bid and ask prices, respectively. $\text{Log}(\text{Time_from_confirmation})$ is the natural log of the number of minutes since the most recent block confirmation. The indicator $\text{Post}[m, n]$ is equal to one if the most recent block confirmation is between m and n minutes ago, and zero otherwise. Reset fixed effects capture observations between difficulty resets, which are 2016 blocks away from each other. DOW and HOD capture the day-of-the-week and hour-of-the-day fixed effects, respectively. Standard errors are clustered at the reset level with the corresponding t-statistics reported in the brackets below. *, **, or *** indicates significance level of 10%, 5%, or 1%, respectively.

Panel A: Time from the most recent block confirmation			
VARIABLES	(1) Ordimb	(2) Return	(3) Return_ba
$\text{Log}(\text{Time_from_confirmation})$	0.951*** [12.84]	0.102*** [16.20]	0.102*** [15.85]
Constant	-1.765*** [-12.59]	-0.183*** [-15.31]	-0.182*** [-14.99]
Observations	3,585,010	3,585,010	3,585,010
R-squared	0.017	0.000	0.000
Reset FE	Yes	Yes	Yes
Panel B: Time from the most recent block confirmation by group			
VARIABLES	(1) Ordimb	(2) Return	(3) Return_ba
$\text{Post}[1, 5]$	-1.056*** [-10.17]	-0.112*** [-5.26]	-0.114*** [-5.63]
$\text{Post}[6, 10]$	0.709*** [8.47]	0.073*** [2.70]	0.069*** [2.65]
$\text{Post}[11, 15]$	1.256*** [11.64]	0.120*** [3.92]	0.121*** [4.14]
$\text{Post}[16, 20]$	1.350*** [10.18]	0.136*** [4.94]	0.132*** [4.85]
$\text{Post}[>21]$	1.289*** [8.50]	0.136*** [5.38]	0.134*** [5.64]
Constant	-0.157*** [-2.78]	-0.007 [-0.35]	-0.005 [-0.27]
Observations	3,585,011	3,585,011	3,585,011
R-squared	0.017	0.000	0.000
Reset FE	Yes	Yes	Yes

Table 6: Exchange flows and post-confirmation effect

This table reports the regressions of minute-by-minute Bitcoin trading and returns on block confirmation events with heterogeneous flows. The sample covers from 2018 to 2024 and the Bitcoin to USDT trading on Binance in the same period. The dependent variables include the market order imbalance (Ordimb) in percentage and the minute return based on closing price and bid-ask price midpoint in basis points. The bid and ask prices are from the last pair of transactions of each minute executed on bid and ask prices, respectively. Post is an indicator that equals one if the most recent block confirmation occurred within five minutes, and zero otherwise. Large_inflow (Large_outflow) is an indicator that equals one if the inflow (outflow) associated with the block is higher than the 80 percentile of the previous reset cycle, and zero otherwise. The control variable, Log_wait, is the natural log of the wait time preceding the block in minute. Reset fixed effects capture observations between difficulty resets, which are 2016 blocks away from each other. DOW and HOD capture the day-of-the-week and hour-of-the-day fixed effects, respectively. Standard errors are clustered at the reset level with the corresponding t-statistics reported in the brackets below. *, **, or *** indicates significance level of 10%, 5%, or 1%, respectively.

VARIABLES	(1) Ordimb	(2) Return	(3) Return.ba
Post×Large_inflow	-0.306** [-2.47]	-0.185*** [-4.23]	-0.189*** [-4.32]
Post×Large_outflow	0.173 [1.39]	0.009 [0.24]	0.011 [0.28]
Post	-1.923*** [-12.55]	-0.166*** [-10.98]	-0.166*** [-10.97]
Large_inflow	-0.103 [-1.06]	0.059*** [2.98]	0.062*** [3.12]
Large_outflow	-0.381*** [-4.09]	0.019 [0.91]	0.017 [0.79]
Log_wait	0.181*** [6.11]	0.019*** [3.27]	0.018*** [3.22]
Constant	0.497*** [7.53]	0.032*** [2.78]	0.033*** [2.84]
Observations	3,585,011	3,585,011	3,585,011
R-squared	0.017	0.000	0.000
Reset FE	Yes	Yes	Yes
DOW FE	Yes	Yes	Yes
HOD FE	Yes	Yes	Yes

Table 7: Performance of confirmation-effect strategies

This table reports the before-cost the after-cost performance of strategies based on confirmation effect. The sample covers from 2018 to 2024 and the Bitcoin to USDT trading on Binance in the same period. Panel A and B report the average before-cost and after-cost daily returns (in percentage) of the strategies in our full sample as well as by calendar year. The long strategy invests only in periods other than the post-confirmation periods. To account for transaction costs, this long strategy buys at the ask prices and sells at the bid prices. The short strategy involves borrowing and shorting Bitcoin in the post-confirmation periods. To account for transaction costs, this short strategy borrows at the margin interest rate, sells at the bid prices and buys back at the ask prices. The post-confirmation periods are defined as the minutes where the most recent block confirmation occurred within five minutes. The bid and ask prices are from the last pair of transactions of each minute executed on bid and ask prices, respectively. The Sharpe ratios (SR) are annualized. Standard errors are clustered at the reset level with the corresponding t-statistics reported in the brackets below. *, **, or *** indicates significance level of 10%, 5%, or 1%, respectively.

Panel A. Before-cost average daily returns									
		Full sample	2018	2019	2020	2021	2022	2023	2024
Buy-and-hold	Daily return	0.139*	-0.323	0.246	0.466**	0.218	-0.225	0.309**	0.255*
	t-stat	[1.92]	[-1.31]	[1.31]	[2.26]	[0.98]	[-1.29]	[2.35]	[1.76]
	SR	0.690	-1.381	1.255	2.240	0.981	-1.317	2.291	1.619
Long	Daily return	0.827***	0.625***	0.601***	0.907***	1.272***	0.474***	0.906***	0.994***
	t-stat	[13.48]	[2.88]	[3.95]	[5.57]	[6.39]	[3.25]	[8.40]	[8.03]
	SR	5.108	3.025	3.883	5.546	6.386	3.216	8.571	7.858
Short	Daily return	0.672***	0.911***	0.342***	0.445***	1.016***	0.690***	0.590***	0.724***
	t-stat	[13.71]	[4.96]	[2.52]	[3.65]	[6.77]	[6.11]	[7.47]	[7.40]
	SR	5.172	5.163	2.445	3.622	6.768	6.066	7.531	7.185
Long + Short	Daily return	1.515***	1.572***	0.955***	1.348***	2.325***	1.173***	1.502***	1.732***
	t-stat	[17.94]	[4.89]	[4.40]	[6.85]	[8.34]	[6.10]	[10.94]	[10.30]
	SR	6.828	5.142	4.356	6.831	8.343	6.077	11.266	10.172
Panel B. After-cost average daily returns									
		Full sample	2018	2019	2020	2021	2022	2023	2024
Long	Daily return	0.144**	-2.307***	-0.623***	0.590***	1.148***	0.253*	0.826***	0.956***
	t-stat	[2.18]	[-10.09]	[-4.14]	[3.57]	[5.77]	[1.74]	[7.76]	[7.70]
	SR	0.800	-10.541	-4.203	3.545	5.770	1.702	7.903	7.532
Short	Daily return	0.037	-1.945***	-0.836***	0.165	0.976***	0.509***	0.530***	0.721***
	t-stat	[0.68]	[-10.12]	[-6.26]	[1.30]	[6.15]	[4.47]	[6.68]	[7.31]
	SR	0.211	-10.586	-6.339	1.276	6.151	4.427	6.705	7.102
Long + Short	Daily return	0.204**	-4.158***	-1.442***	0.749***	2.162***	0.770***	1.361***	1.691***
	t-stat	[2.16]	[-12.48]	[-6.78]	[3.70]	[7.59]	[4.01]	[10.04]	[10.04]
	SR	0.810	-13.038	-6.829	3.684	7.593	3.981	10.322	9.903